Breaking the Power Wall: Chiplet-Based Power Management Architectures for HPC 2.5D/3D Processors

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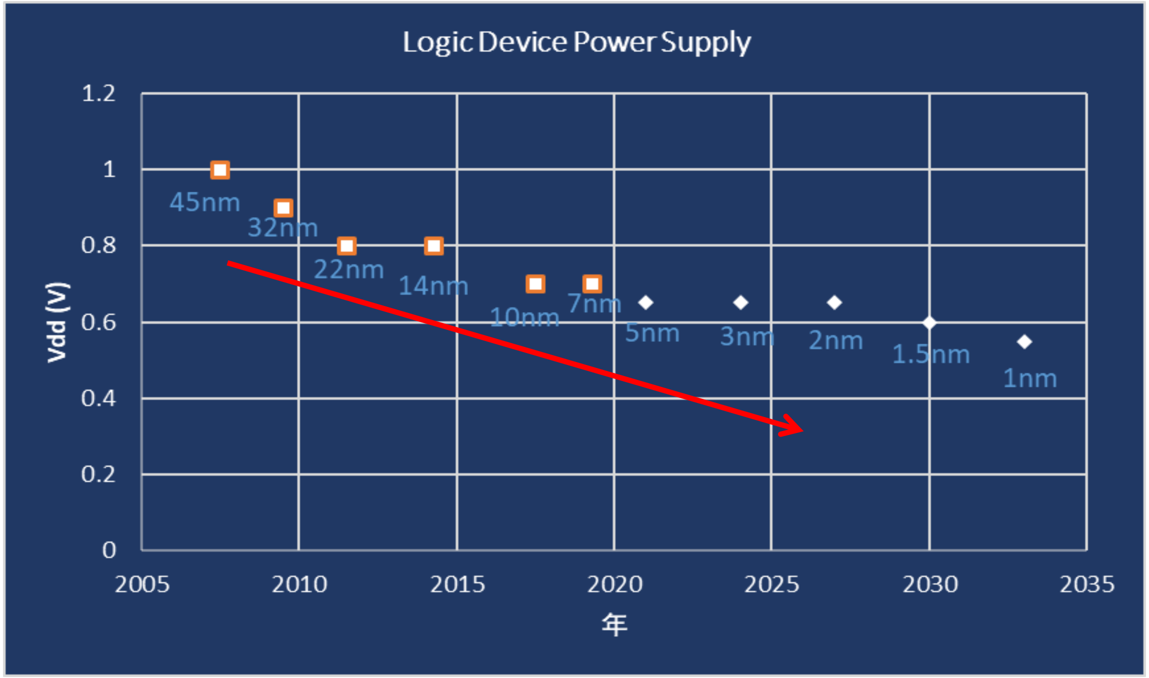
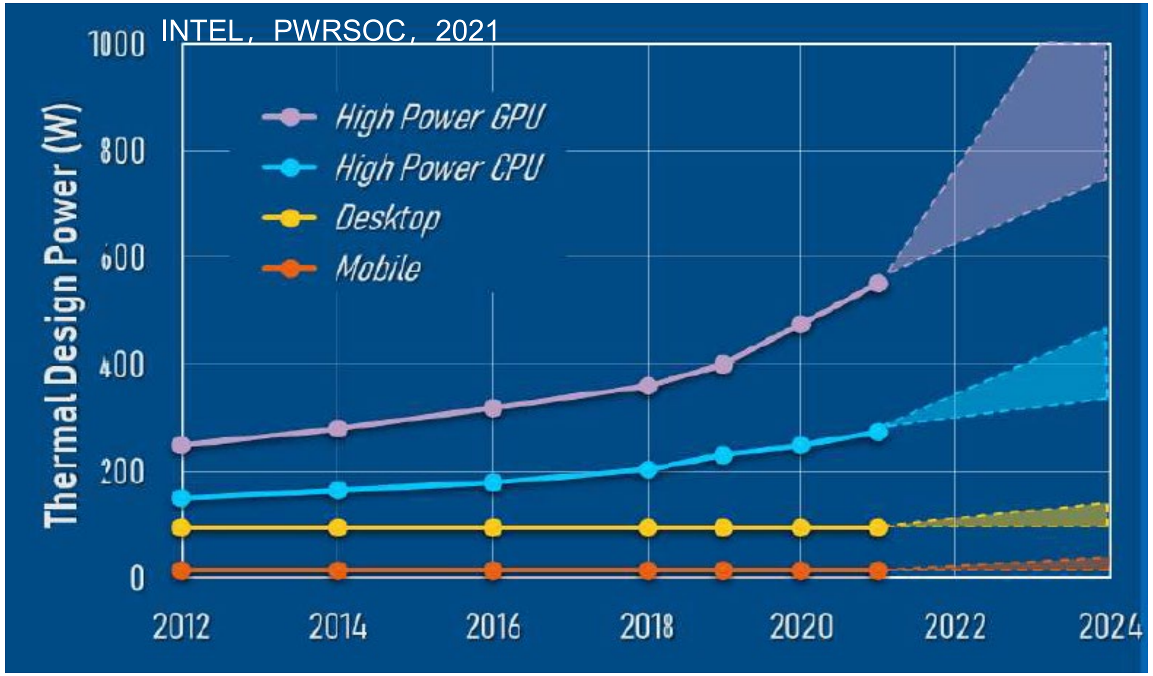
OUTLINE

- **Next-Generation Datacenter and *Power Wall* Bottleneck**
- Integrated Voltage Regulator Chiplet for 2.5D/3D Processors
- Vertical Power Delivery (VPD)
- Power Architecture for HPC Datacenter
- Conclusions

Technology Trend for HPC Processors

CPU Process, Voltage and TDP Benchmark

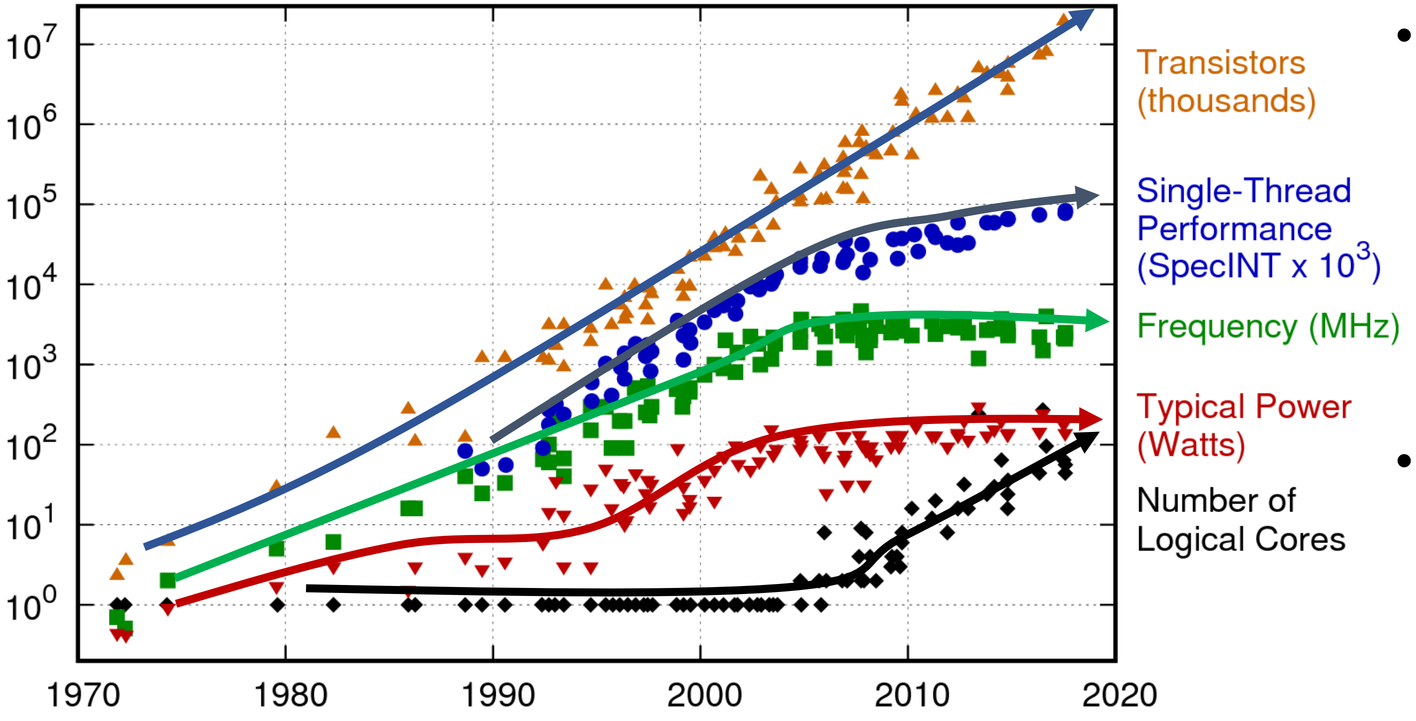




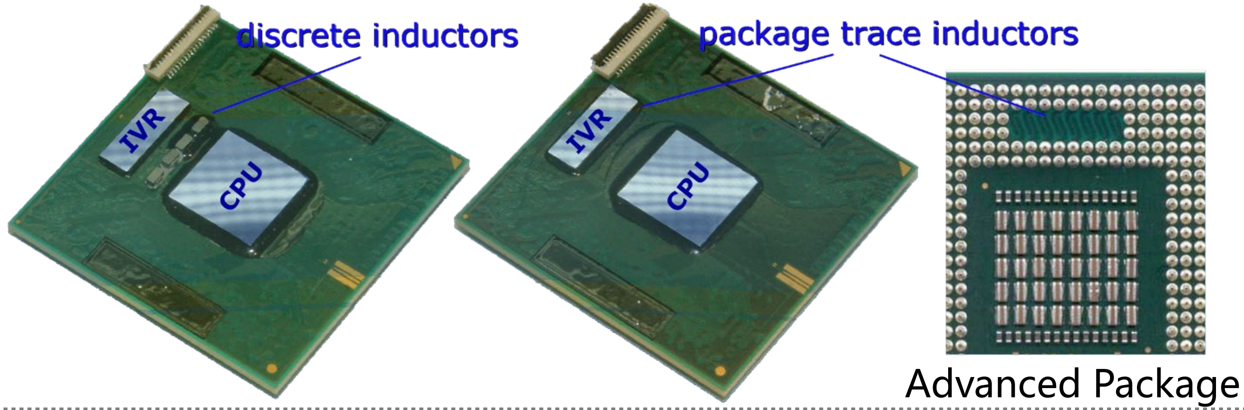
➢**Power management becomes the major bottleneck → *Power Wall***

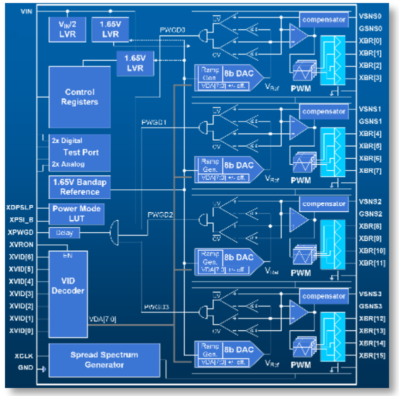
- **High Accuracy**
- **High Current**
- **Low PDN Resistance**
- **Fast di/dt Response**

Motivation-Power Challenges in Future HPC



- **The "Power Wall" Bottleneck in the Post-Moore Era**
  - ➢ **100 Billions of transistors ≠ full perf realization ☹**
  - ➢ **Limited power delivery cannot sustain simultaneous operation of massive gates**
- **New paradigm: Power Delivery is IPC**
  - ➢ **Beyond Moore's Law → delivering more power within limited volume**
  - ➢ **Conventional PD arch: ~1 A/mm² current density, ~90% eff → the last mile to XPUs**





Fully Integrated Voltage Regulator (FIVR)

2D MCM in Intel® Core™

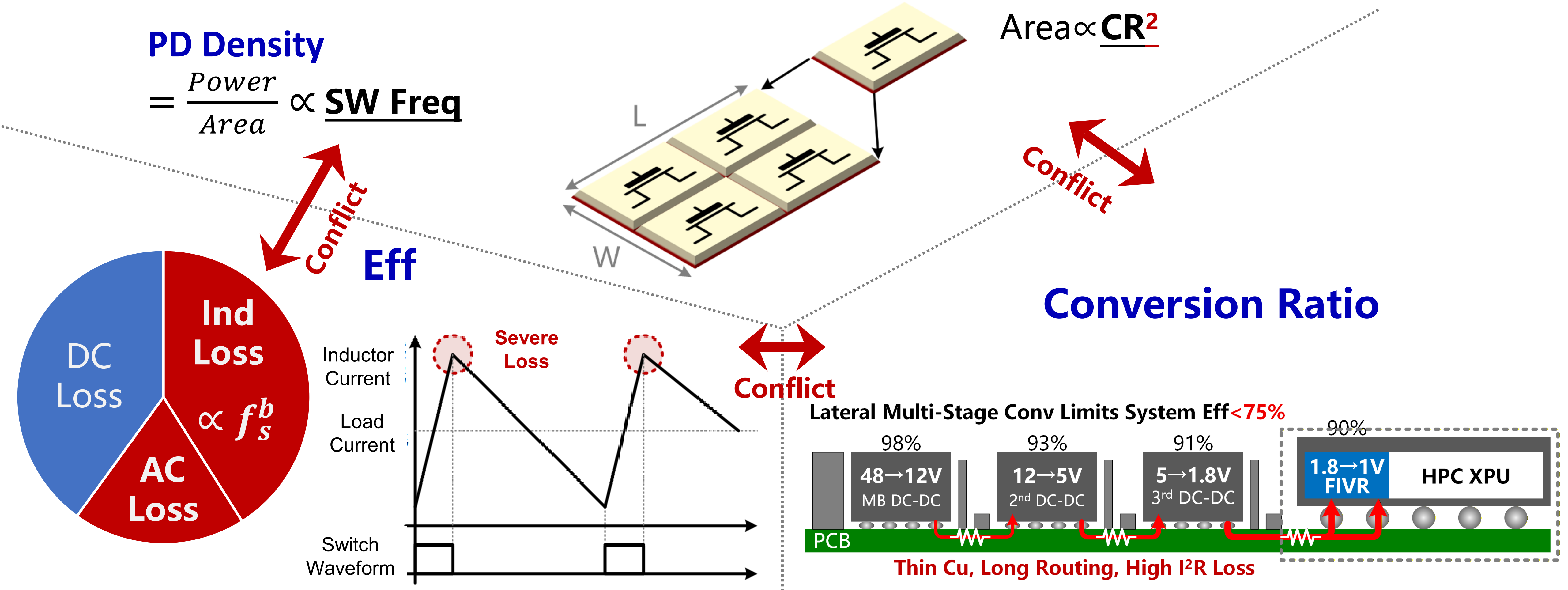
1.3A/mm², 85%

3.3V→1V, 50A I_{max}

State-of-the-Art Performance

Motivation-Tradeoff between Critical Performances

Trade-offs among efficiency, power density, and conversion ratio limit conventional converters to ~1 A/mm², constraining XPU performance scaling.

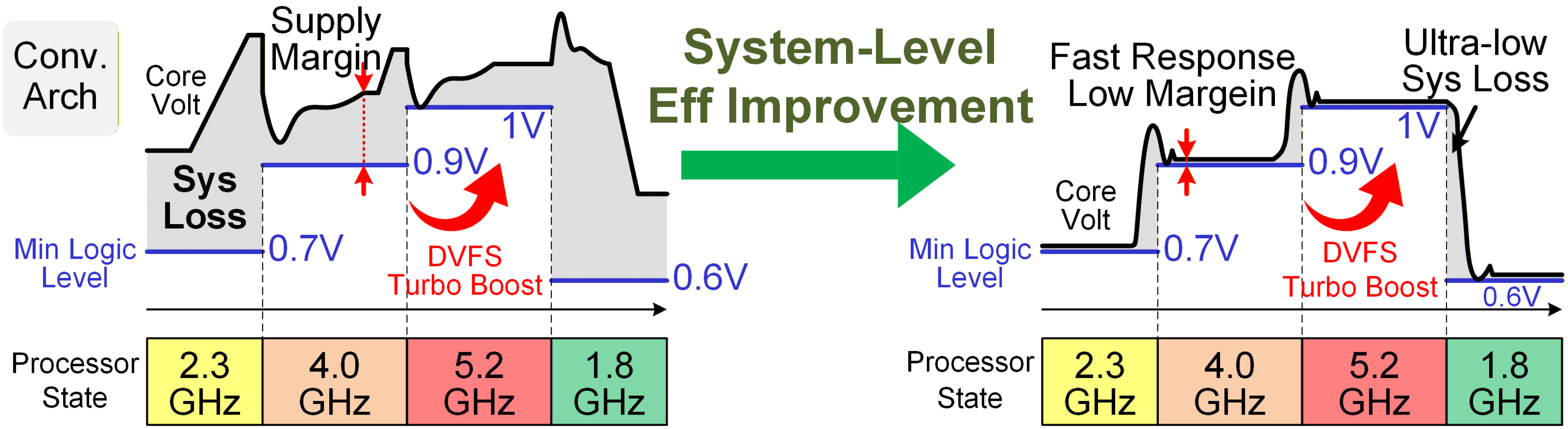


Motivation-Power Management Determines Compute Speed

DVFS demands overcoming low-frequency power delivery limits through fast transient and precise regulation to unlock XPU performance.

$P_{AC\_loss} = C V_{DD}^{2} f$ ➡ $\Delta 0.1V \Rightarrow 30\sim50\%$ ***Loss***

Turbo Boost Technology:
Fast Response ⇄ Core Voltage Boost
⇄ Accelerated Computation



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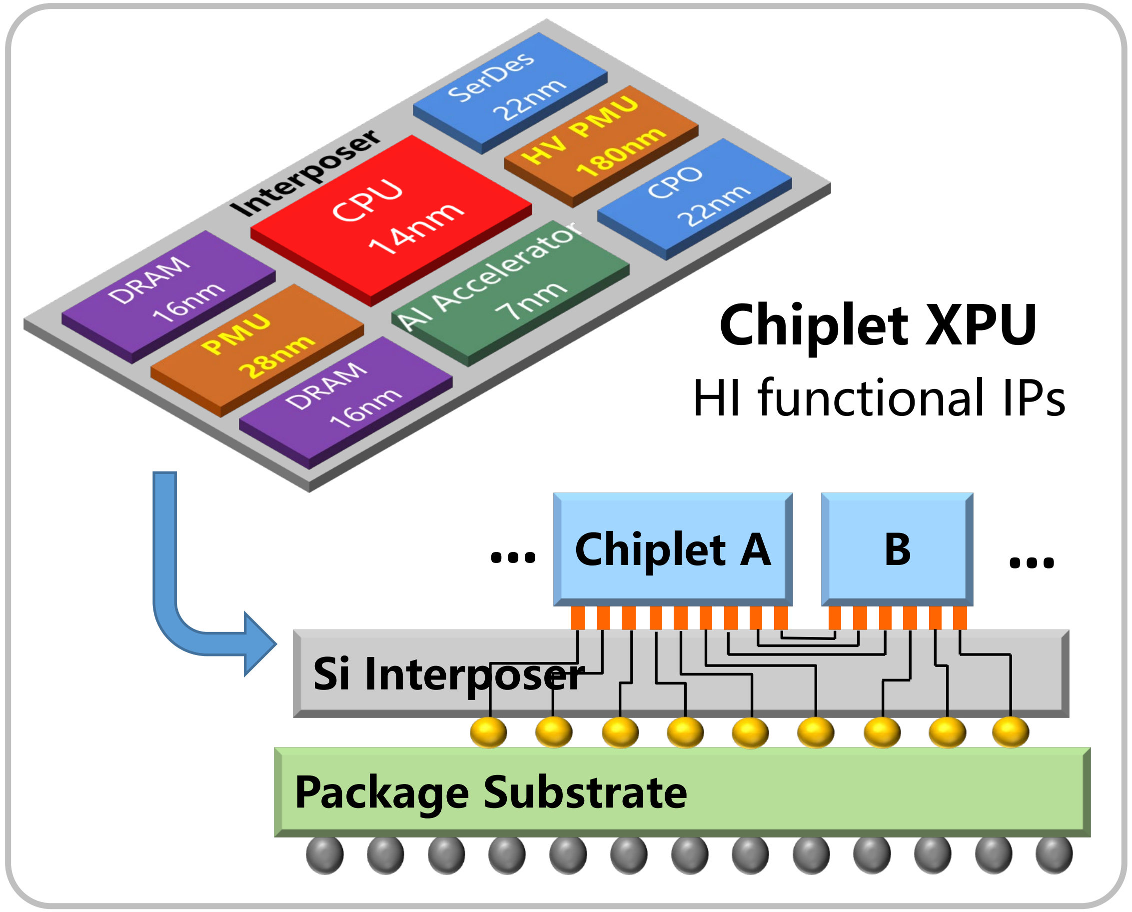
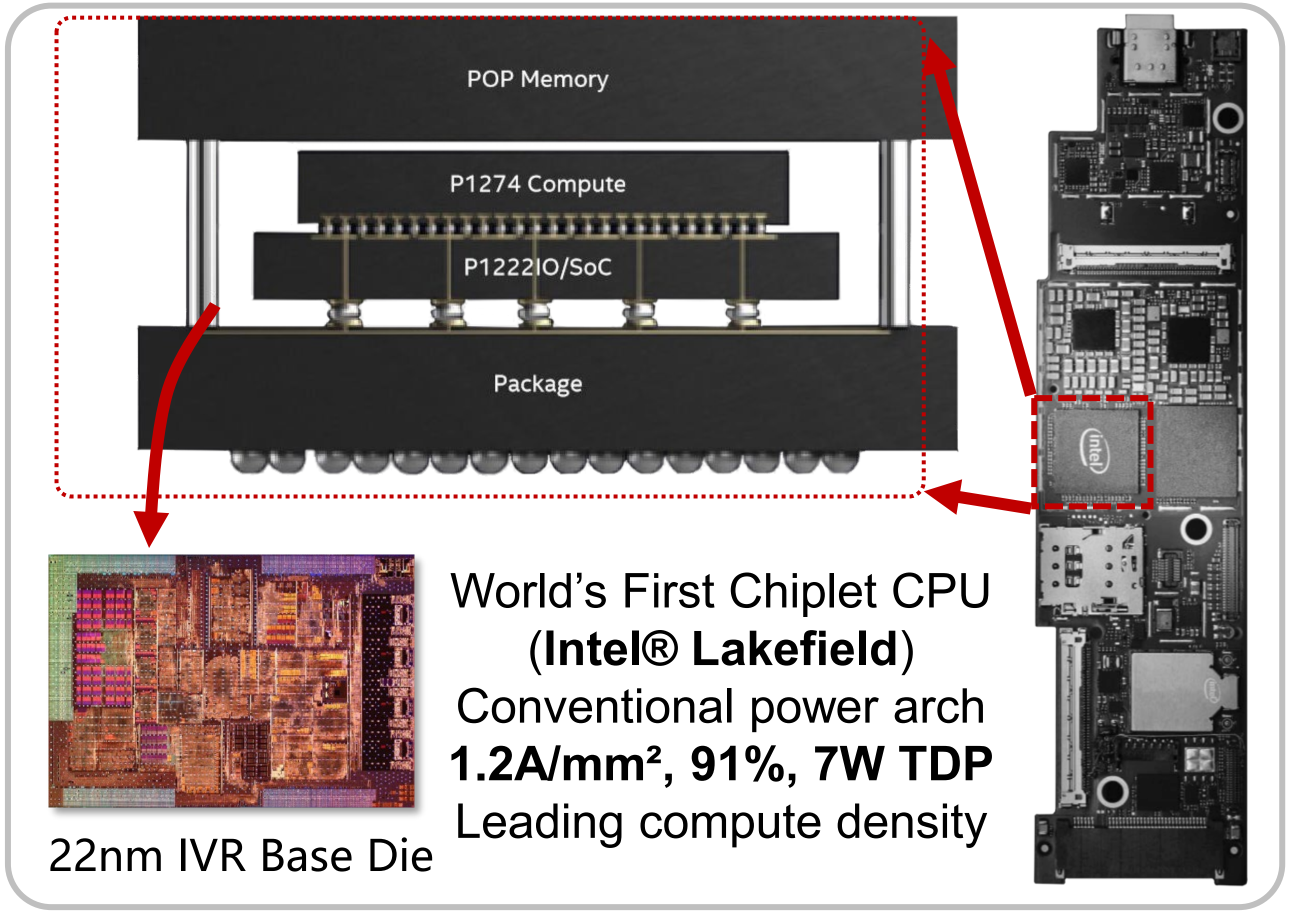
Power Challenge-Interdisciplinary Frontier in Chiplet SoCs

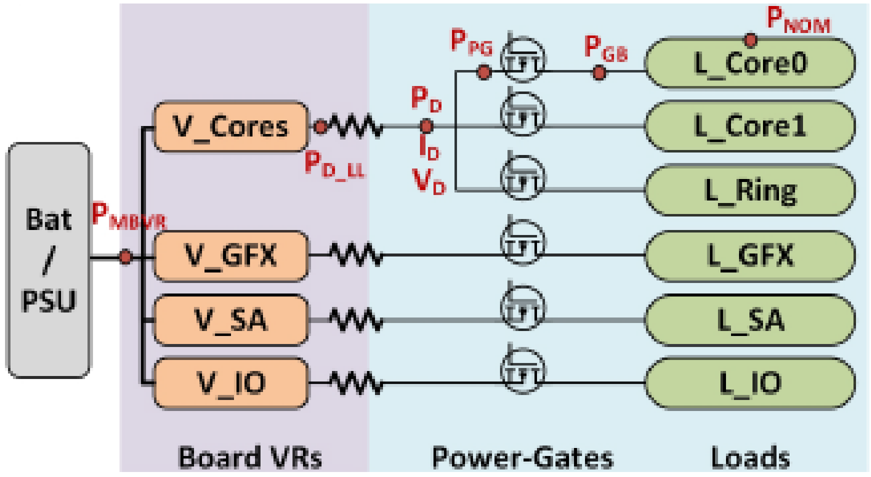
Chiplets enable high performance and agile development in the post-Moore era.

Power management and chiplets together drive future processor performance.





IVR Roadmap for 2.5D/3D Processors



(MBVR)

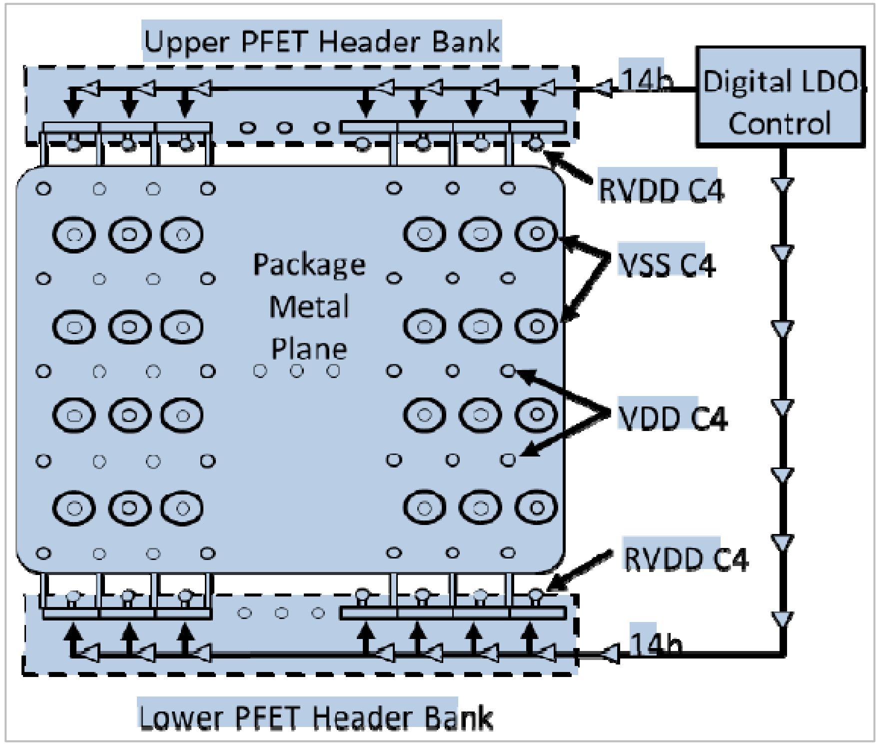
Conventional Motherboard-based Power Delivery Solution

One Big Power Domain >500W

AMD: Lateral Power Supply + Per-Core Voltage

LDO

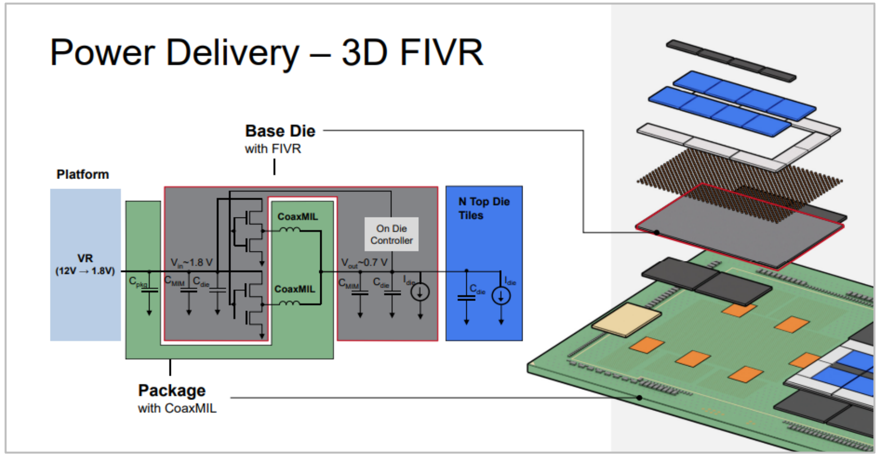
ISSCC 2020



Integrated Buck

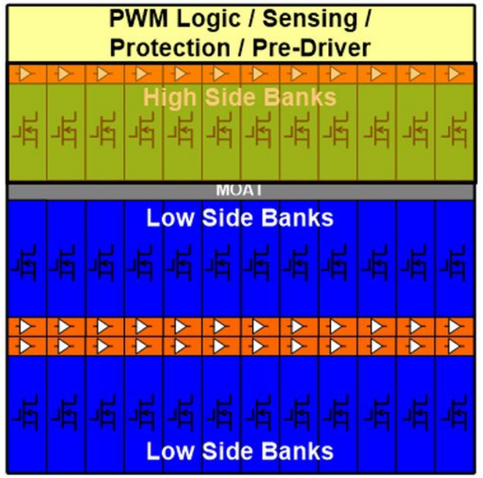
Intel: Lateral Power Supply + Per-Core Voltage

ISSCC 2022



Four Technology Trends for Advanced PD

MOSFET Process

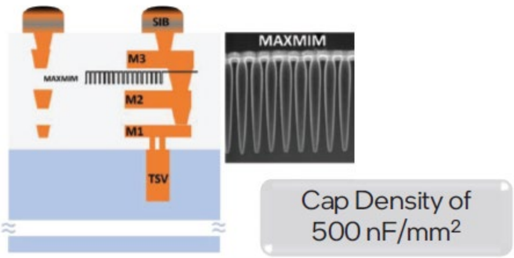


55nm BCD

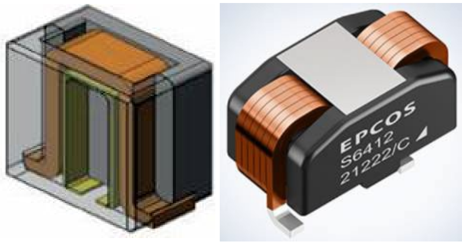
90nm BCD

IPD

MAXMIM (MIM with 1um Trench)

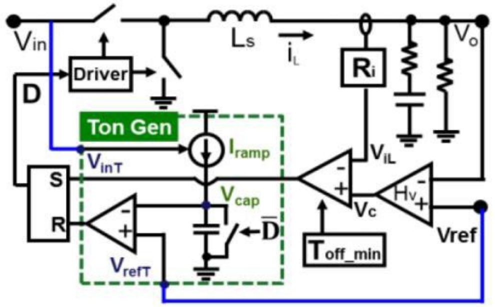


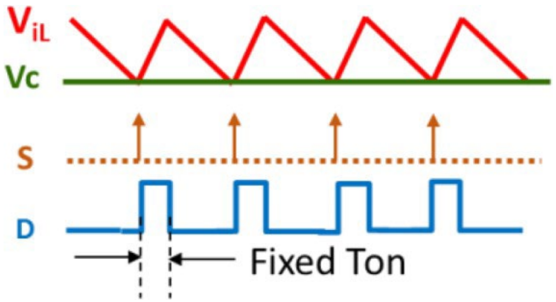
High-Density Cap



TLVR and Inductor

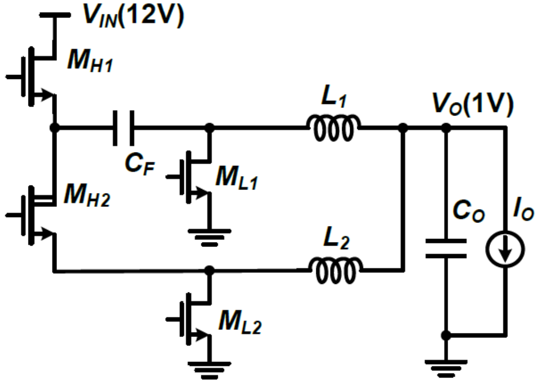
Control

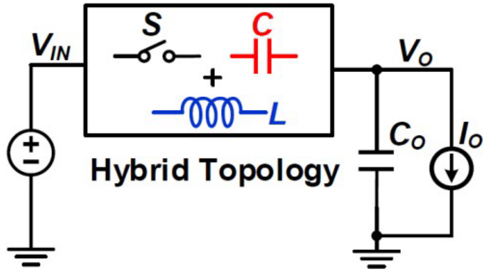




Optimized Transient

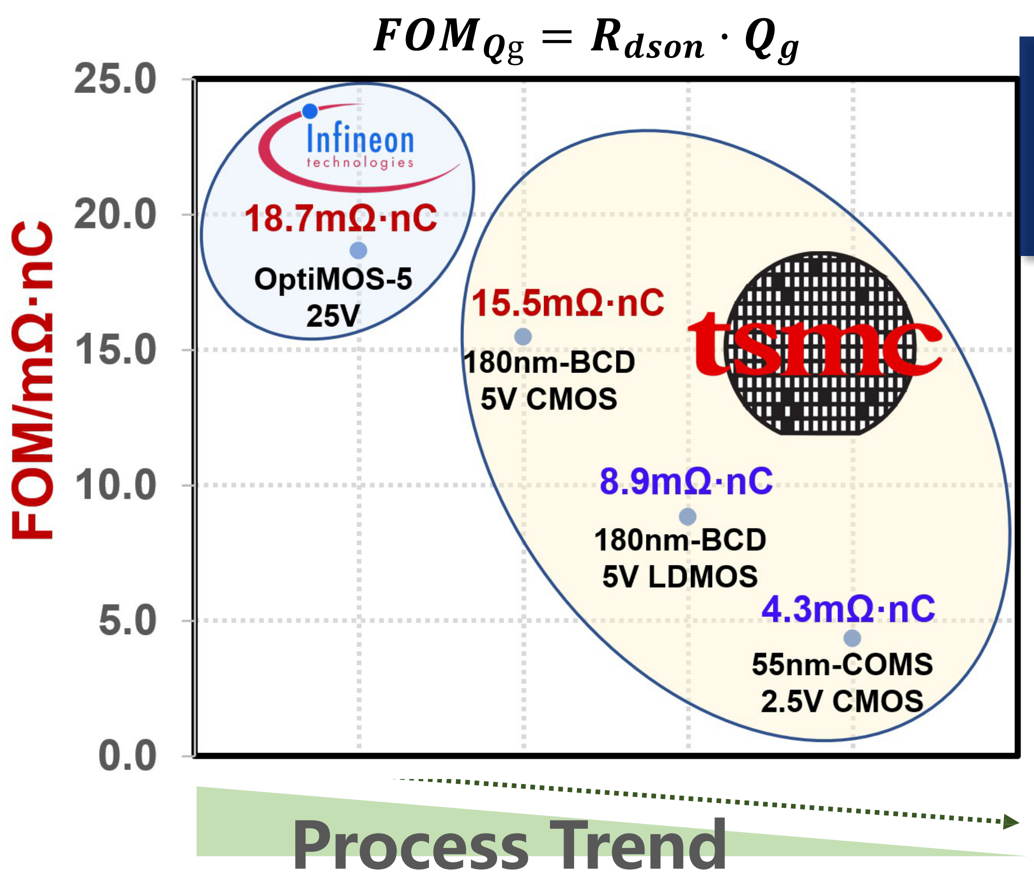
Topology





Hybrid Arch

MOSFET Process Also Drive the Integrated Power



Trench-MOS

Infineon

Trench MOSFET
OptiMOS™ 5 25V → OptiMOS™ 7 15V

R_{DSon}($FOMQ_g$) ↓ ~**30%**, $FOMQ_{OSS}$ ↓ **~40%**

BCD

ST life.augmented, SMIC, ANALOG DEVICES

90nm BCD

MPS Monolithic Power Systems

55nm BCD gen6

R_{DSon} ↓~**25%**, CMOS area ↓↓

CMOS

intel

7/22nm FinFet CMOS node

tsmc

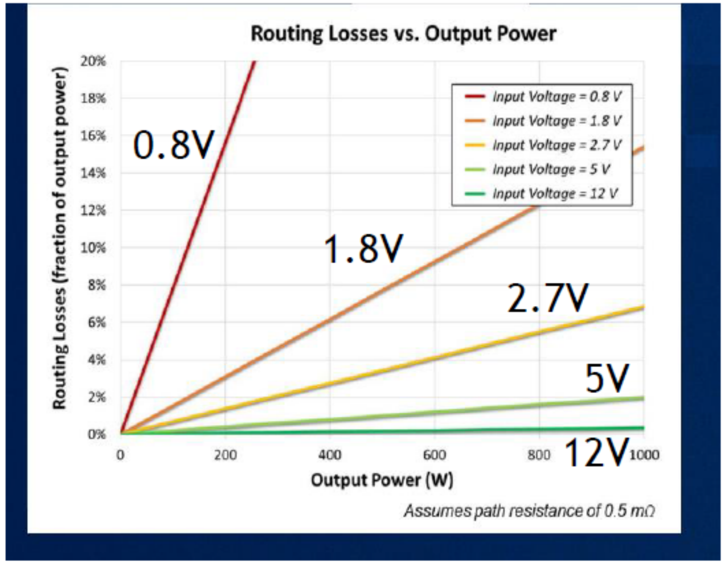
12nm FinFet CMOS

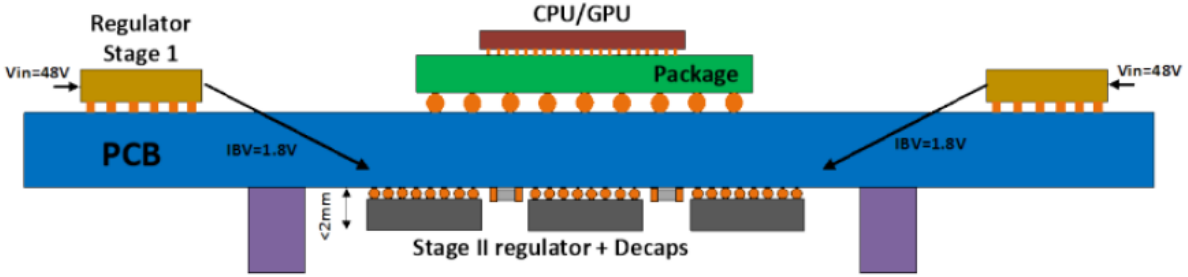
Advanced Node ➔ Better FOM ➔ Higher Freq + Density

Requirements for On-Package Voltage Regulator (OPVR)

- Higher voltage eliminates routing losses → >5V keeps losses to <2%
- Low z-height (<2mm) → Low profile Inductors → Higher IVR frequency
- Minimal cooling on backside → High efficiency of IVR

Source: K Radhakrishnan POWERSOC 2021





Source: Sudhir Kudva, APEC 2024

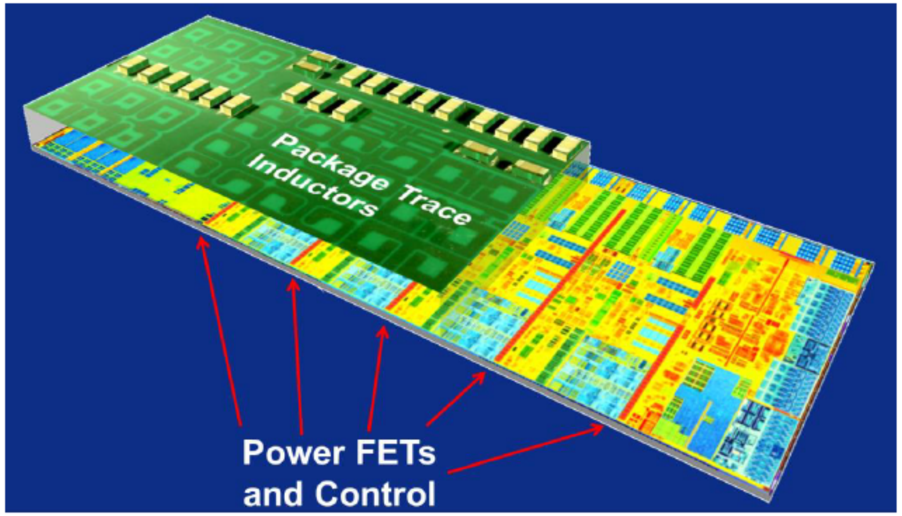
VPD: Vertical Power Delivery
OPVR: On-package Voltage Regulator

Ref.: Harish K Krishnamurthy, “A 5.4V-Vin, 9.3A/mm2, 10MHz Buck IVR chiplet in 55nm BCD featuring self-timed bootstrap and same-cycle ZVS control,” VLSI 2024.

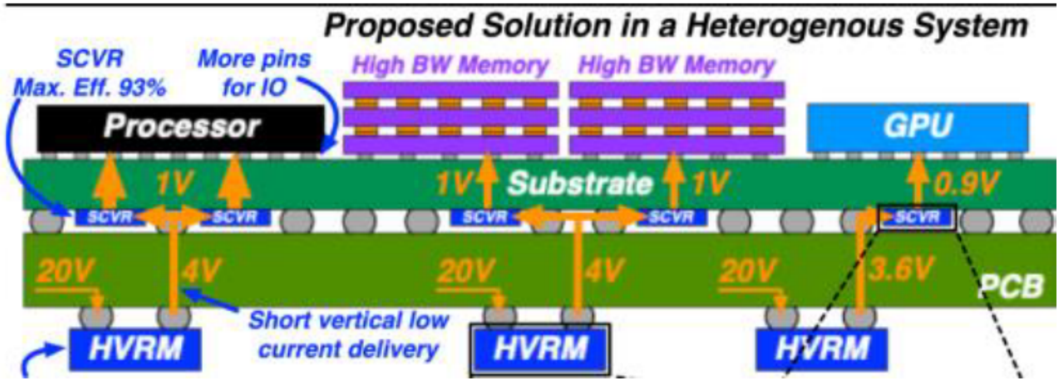
Current State of the Art FIVR

- FIVR has demonstrated high efficiency and high current density
  - Input voltage however is limited to 1.8V
- Hybrid Switched Capacitors promise high efficiency for large VCR
  - Maximum Current densities are limited (<0.5A/mm2)

Source: Edward (Ted) Burton APEC 2014



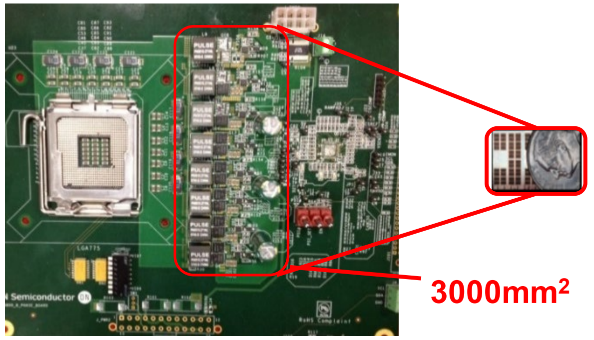
Source: Casey Hardy, ISSCC 2023



VCR: Voltage Conversion Ratio

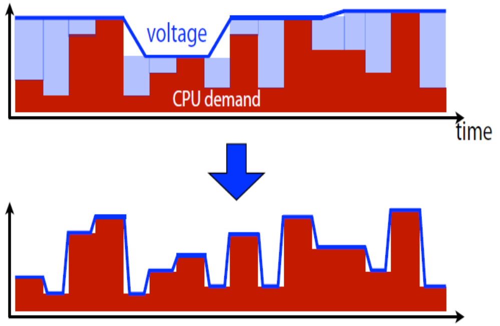
Ref.: Harish K Krishnamurthy, "A 5.4V-Vin, 9.3A/mm2, 10MHz Buck IVR chiplet in 55nm BCD featuring self-timed bootstrap and same-cycle ZVS control," VLSI 2024.

Advantages of Fully Integrated Voltage Regulator (FIVR)

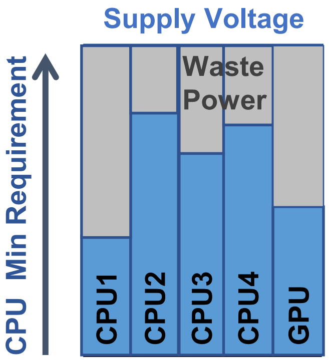


- 20 Independent PMU
- 16 phases
- Total 160 Coupled INDs

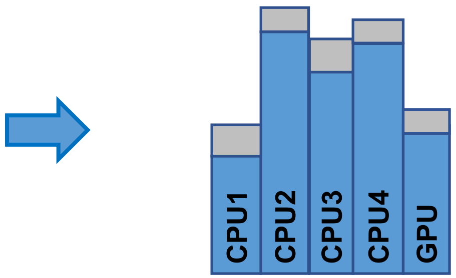
➢ **Reduce bulky PMU area & height**



➢ **Faster transients**

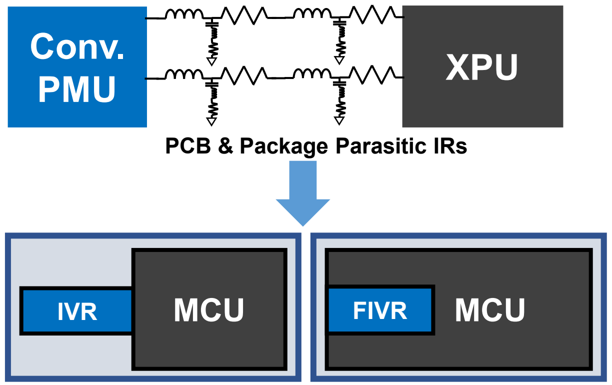


Constant Supply



Adaptive Supplies for Each Core

➢ **Finer voltage regulation**



➢ **Smaller IR losses**

Advantages

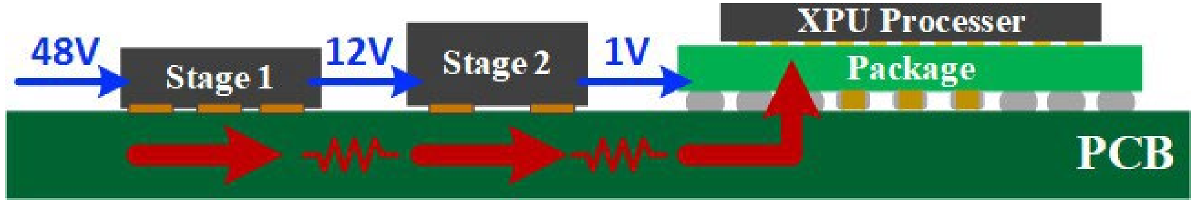
Efficiency

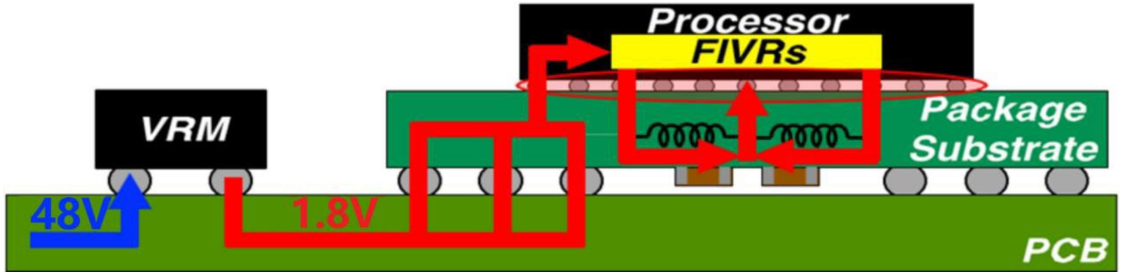
Reliability

Weight

Form Factor

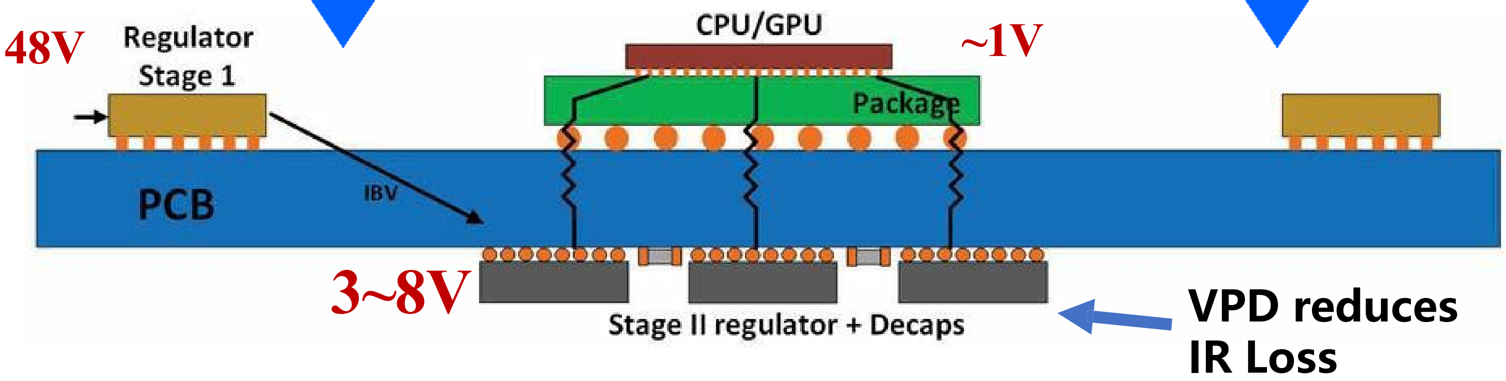
Power Supply Roadmap for HPC 2.5D/3D Processors





High 12V IBC
Limits Density & Speed

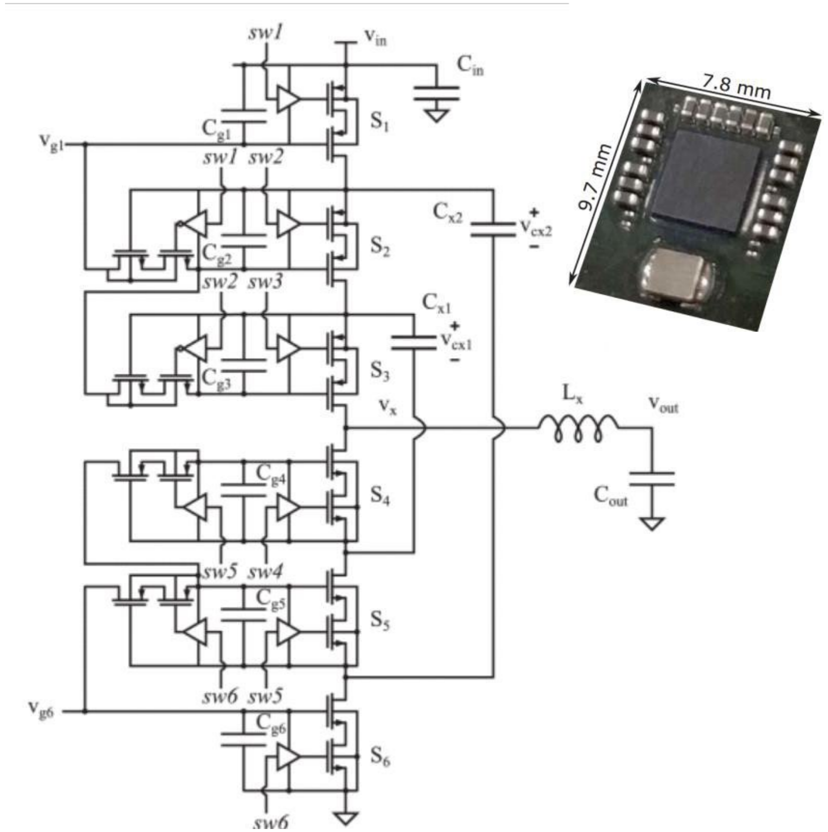
Low 1.8V IBC
Limits Eff & max power



- Max Pout
- Density
- Speed

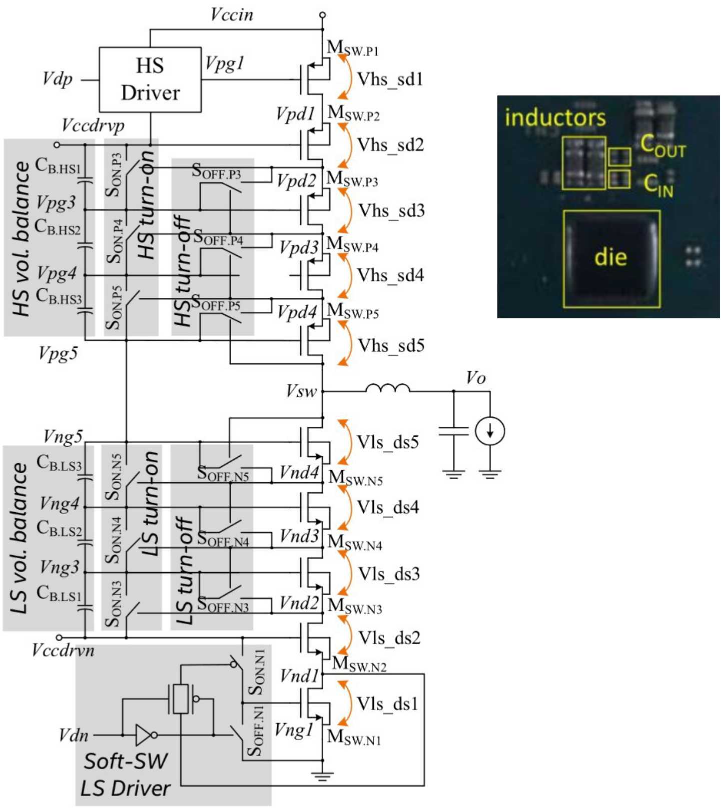
48V Input + 3–8V Bus: Key Two-Stage PD Arch

Trend Toward Higher Voltage in IVR Chiplets



Intel, ISSCC,2019 8.1

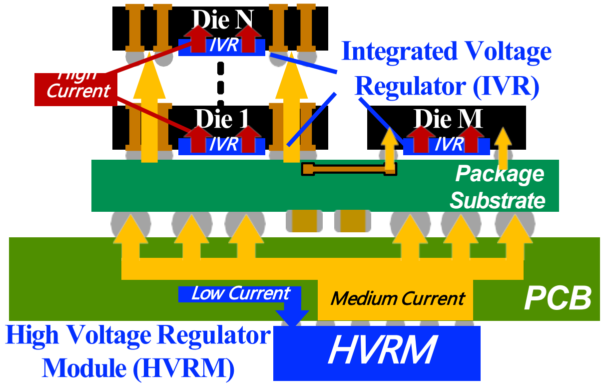
Multi-level



Intel, SSC Letter, 2021

Stacked Device

On-Chip HV Arch

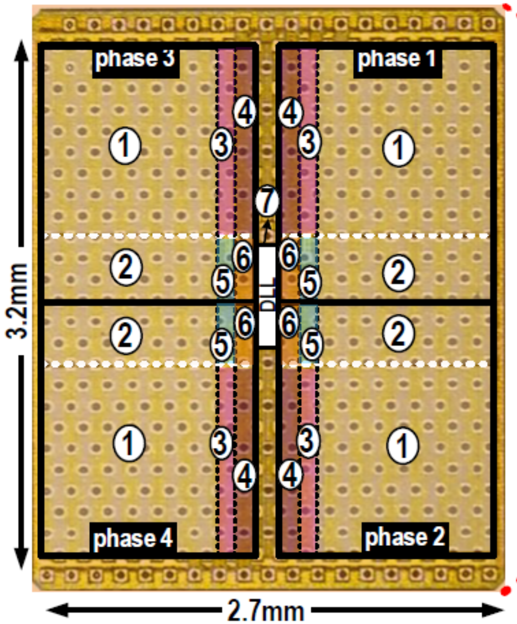


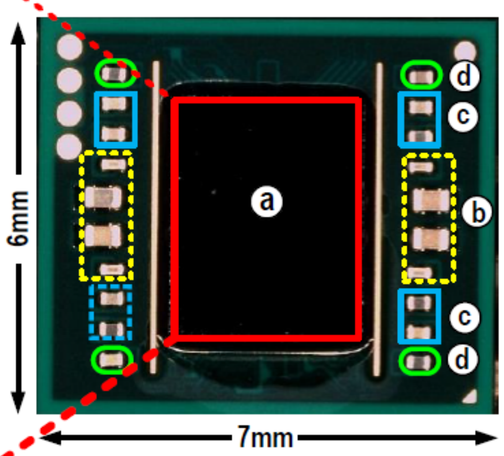
- HV IVR **(>4V)** for on-chip low-voltage, high-current
- **Reduced package current stress → higher XPU power density**
- Fewer power pins → increased **data bandwidth**

Package-Integrated HV Arch Breaking Dynamic and Density Limits

High-Frequency High-Voltage On-Package IVR Chiplets

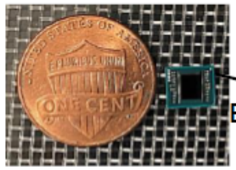
- Die Area: 2.7x3.2mm2 IVR chip
- Package Size: 7x6mm2
- BGA package along with package decoupling and BST capacitors
- Input Voltage Range: 3.8-5.4V
- Output Voltage Range: 0.6-1.8V
- Load Current Range: 0-80A
- # of Phases: 4





① Low-Side Switch [4.86mm²]
② High-Side Switch [1.66mm²]
③ Low-Side ZVS Driver [0.506mm²]
④ Decaps [0.586mm²]
⑤ High-Side Driver [0.115mm²]
⑥ Self-Timed Bootstrap Switch [0.148mm²] + Floating Level-shifter [0.0173mm²]
⑦ Delay-Locked Loop (DLL) [0.066mm²]

ⓐ Fabricated IVR Chip on 55nm BCD
ⓑ Input Voltage Decaps (4x0201+4x01005)
ⓒ Boot-Strap Caps (2x01005 per phase)
ⓓ Driver Voltage Decaps (4x01005)



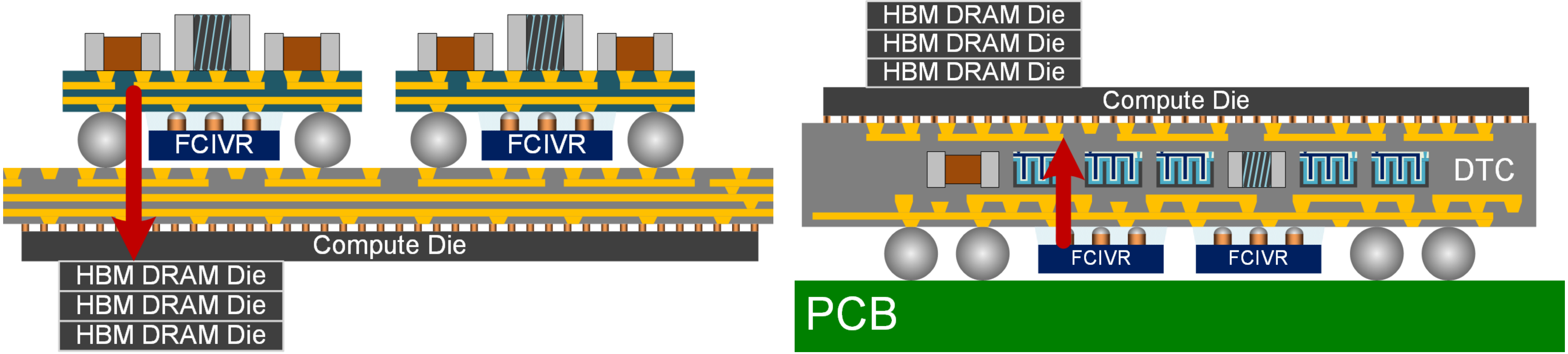
7x6 BGA package

IVR: Integrated Voltage Regulator
BGA: Ball Grid Array

Ref.: Harish K Krishnamurthy, "A 5.4V-Vin, 9.3A/mm2, 10MHz Buck IVR chiplet in 55nm BCD featuring self-timed bootstrap and same-cycle ZVS control," VLSI 2024.

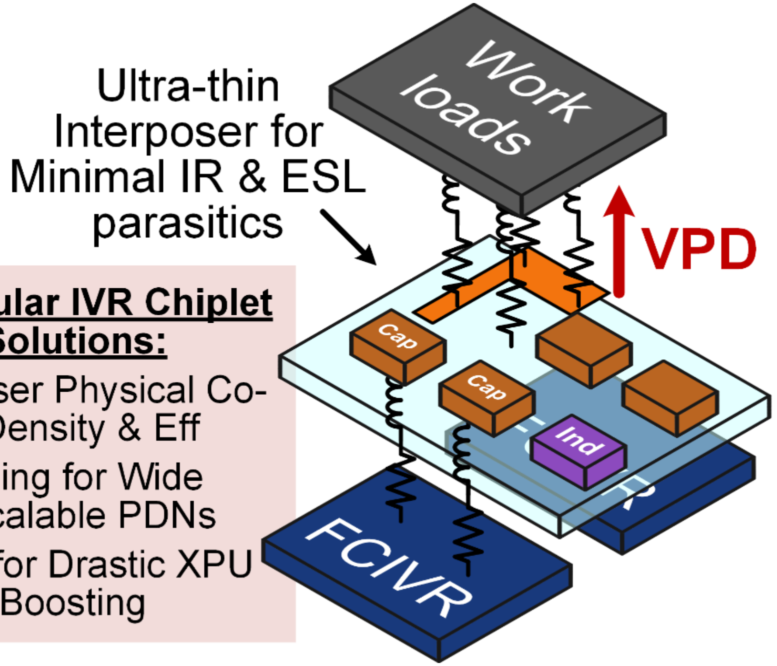
Proposed Integration Architecture

- Flying Capacitor IVR (FCIVR) chiplet can be heterogeneously integrated into 2.5D/3D-ICs using two VPD methods:
  - Attached to the opposite side of the RDL hosting the compute die, operates as an independent power tile w/ IPDs and can be ganged arbitrarily in an array
  - Attached face-up towards the IPDs and compute die embedded in substrate via TSV/TMV technologies, allowing vertical current flow through



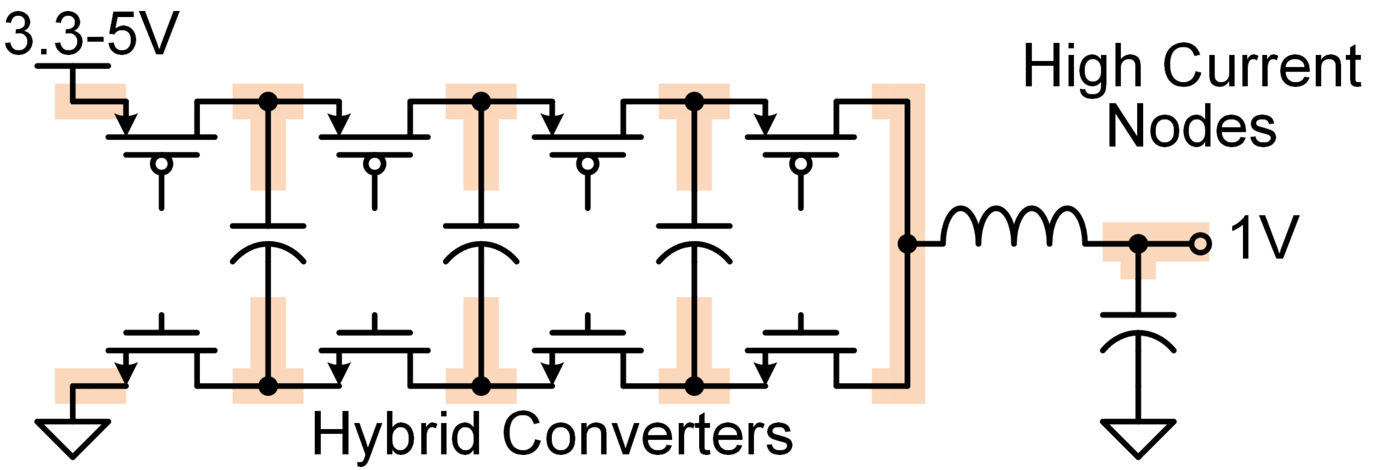
Proposed Circuit Ganging Topology

- Key power management tradeoff for HPC workloads
  - Wide load range vs. high current density
  - High equivalent fsw vs. power conversion efficiency (PCE)
  - Unified voltage domain for drastic di/dt changes and minimal droop



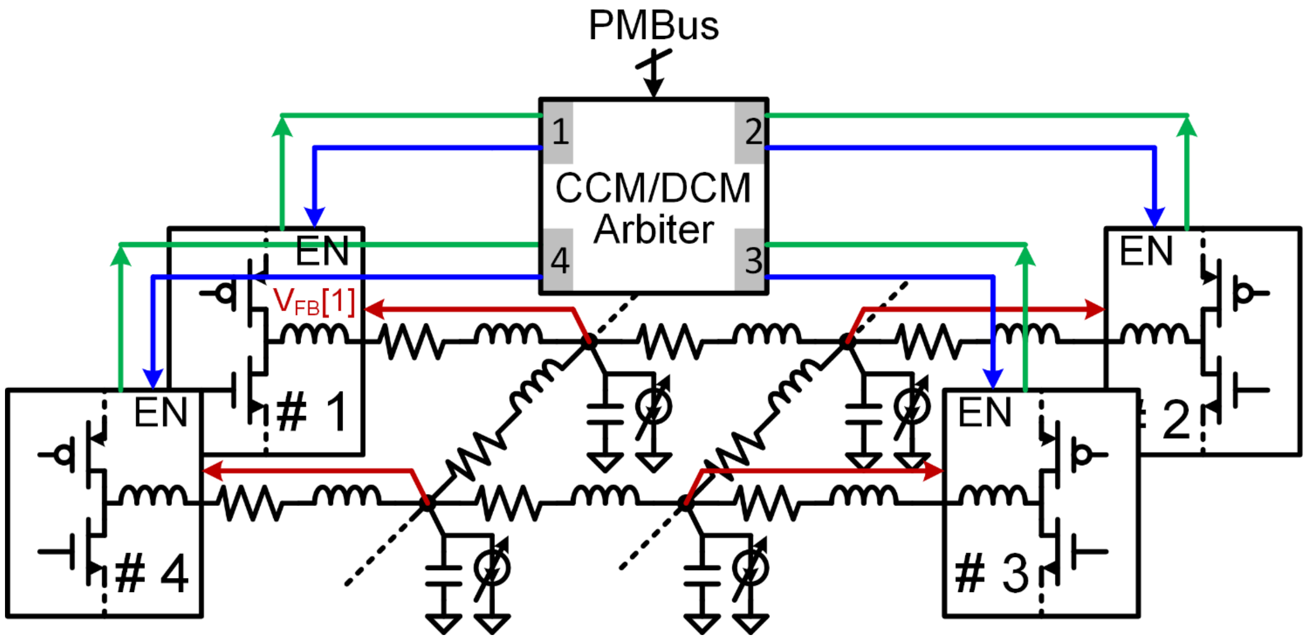
<u>Proposed Modular IVR Chiplet for VPD Solutions:</u>

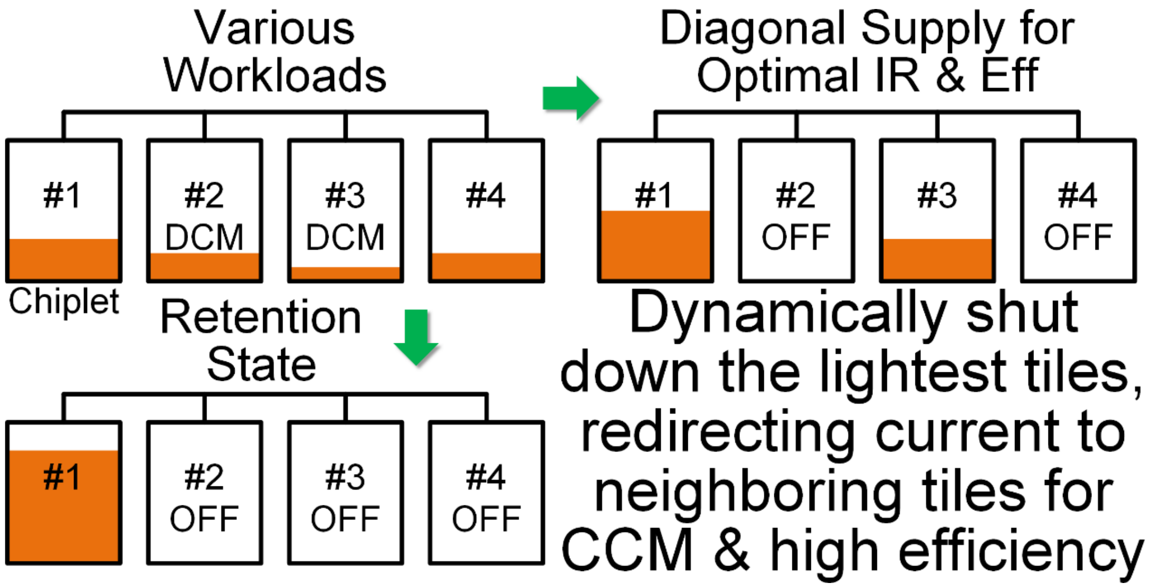
1. Silicon/Interposer Physical Co-Design for High Density & Eff
2. Adaptive Ganging for Wide Load Range & Scalable PDNs
3. Nonlinear Ctrl for Drastic XPU Response & IPC Boosting



Current Steering Mode for Adaptive PDN

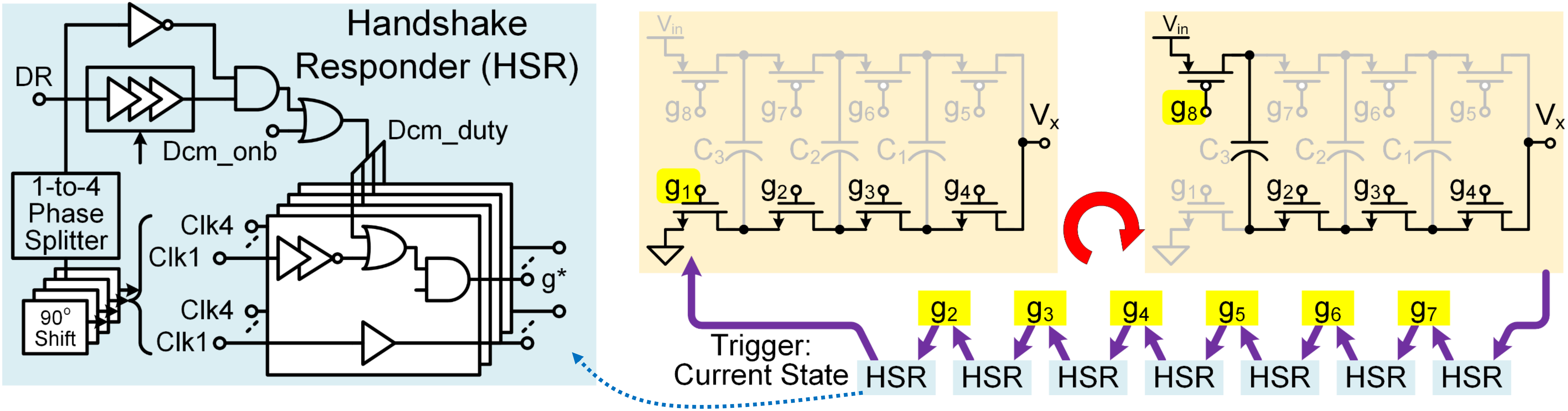
- Ultra-light load conditions for skewed workload allocation
  - Shuts down the lightest tiles and redirects the current to neighboring FCIVRs for CCM conversion, effectively scaling down I_Q and maintaining high PCE
  - CCM/DCM arbiter monitor powertrain and evaluate with tile affinity info





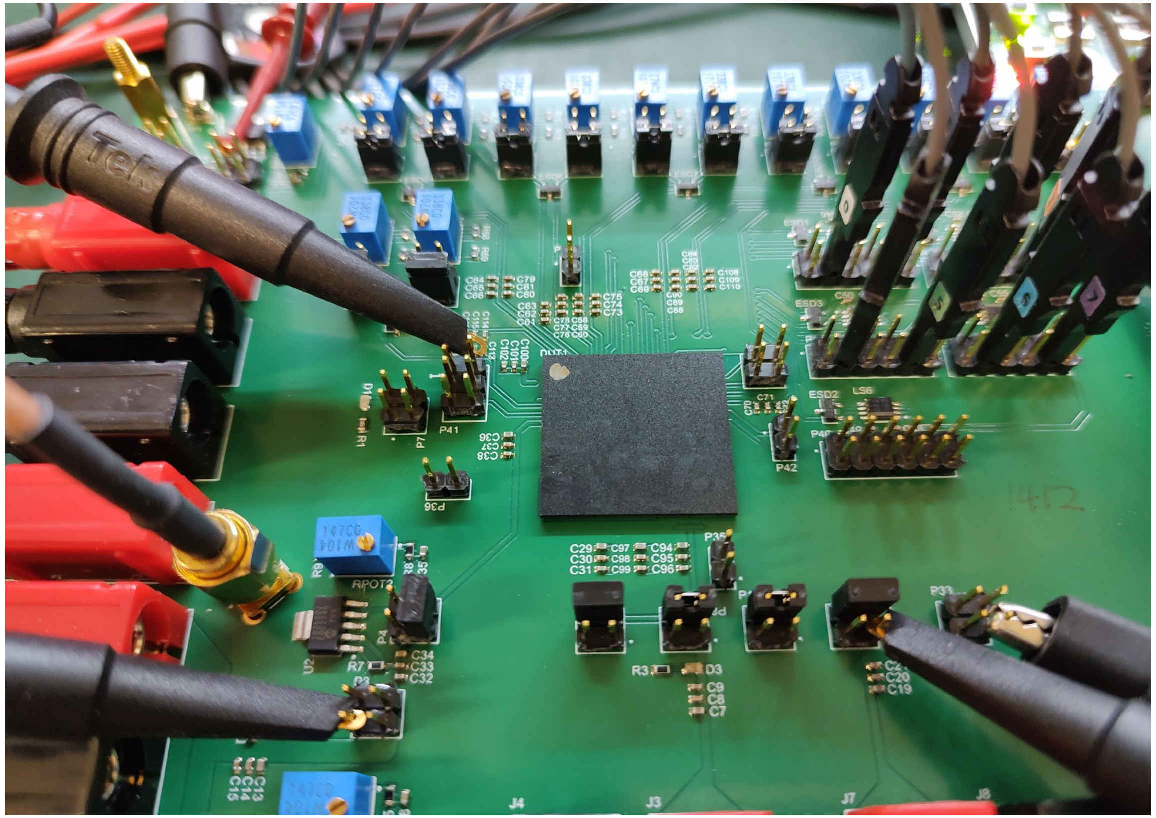
Autonomous Handshake Driving Strategy

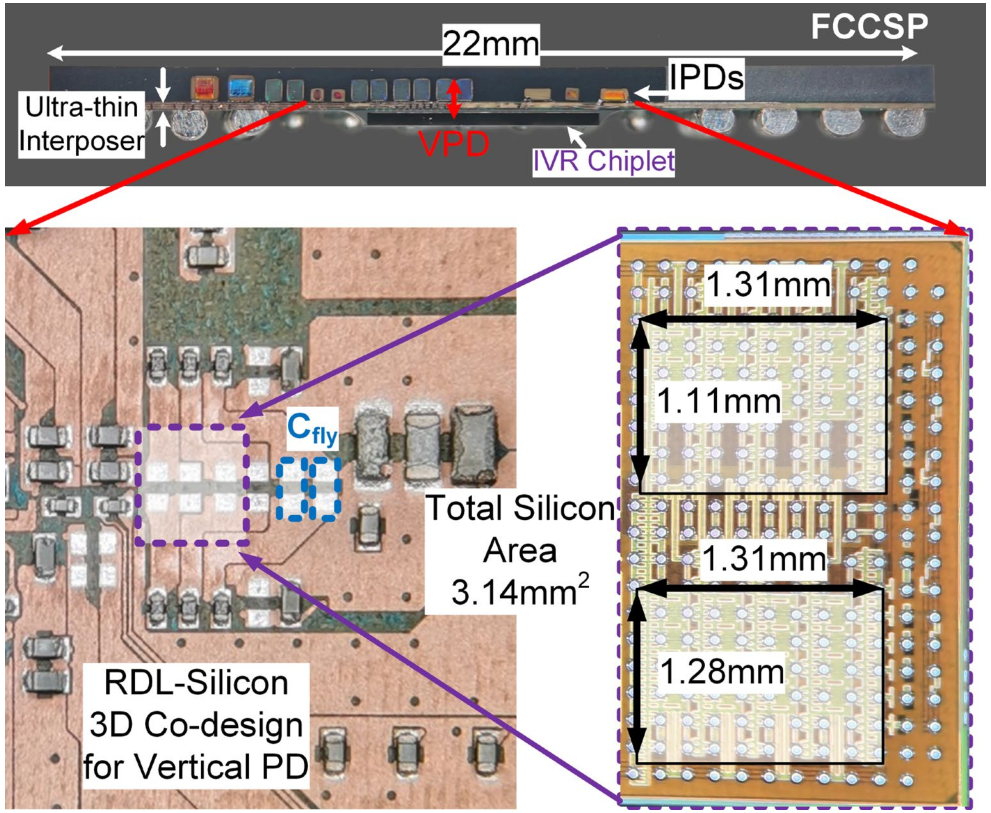
- Proposed mechanism can manage increasingly complex switch network
  - Arrange a logic handshake responder (HSR) to decide the proper shifting trigger sequence under ½ or ¼ conversion ratio settings
  - Scalable for arbitrary number of stages and optimal deadtime for better PCE



System Implementation

- FCIVR VPD chiplet fabricated in 65nm CMOS process. Heterogeneously integrated with an ultra-thin interposer and vertically stacked IPDs
  - 22nH high-Q inductor, three pairs of 4.7μF flying capacitors in FCCSP package

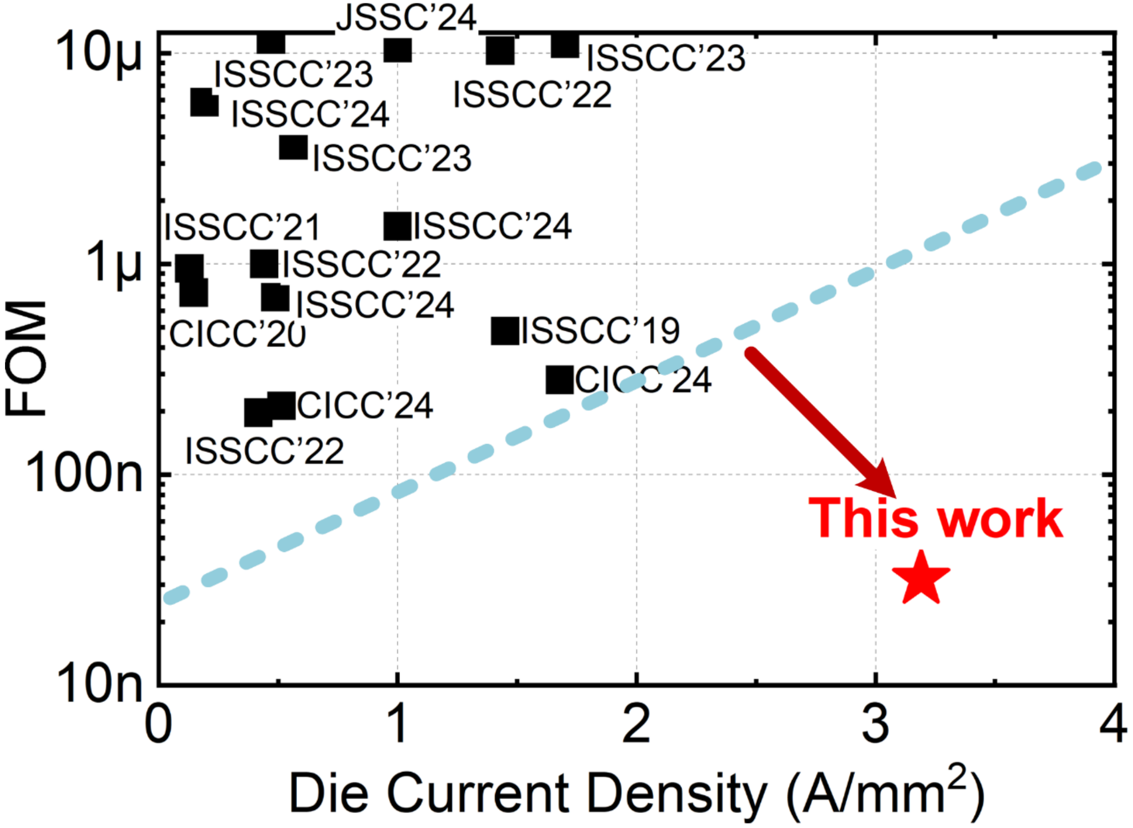




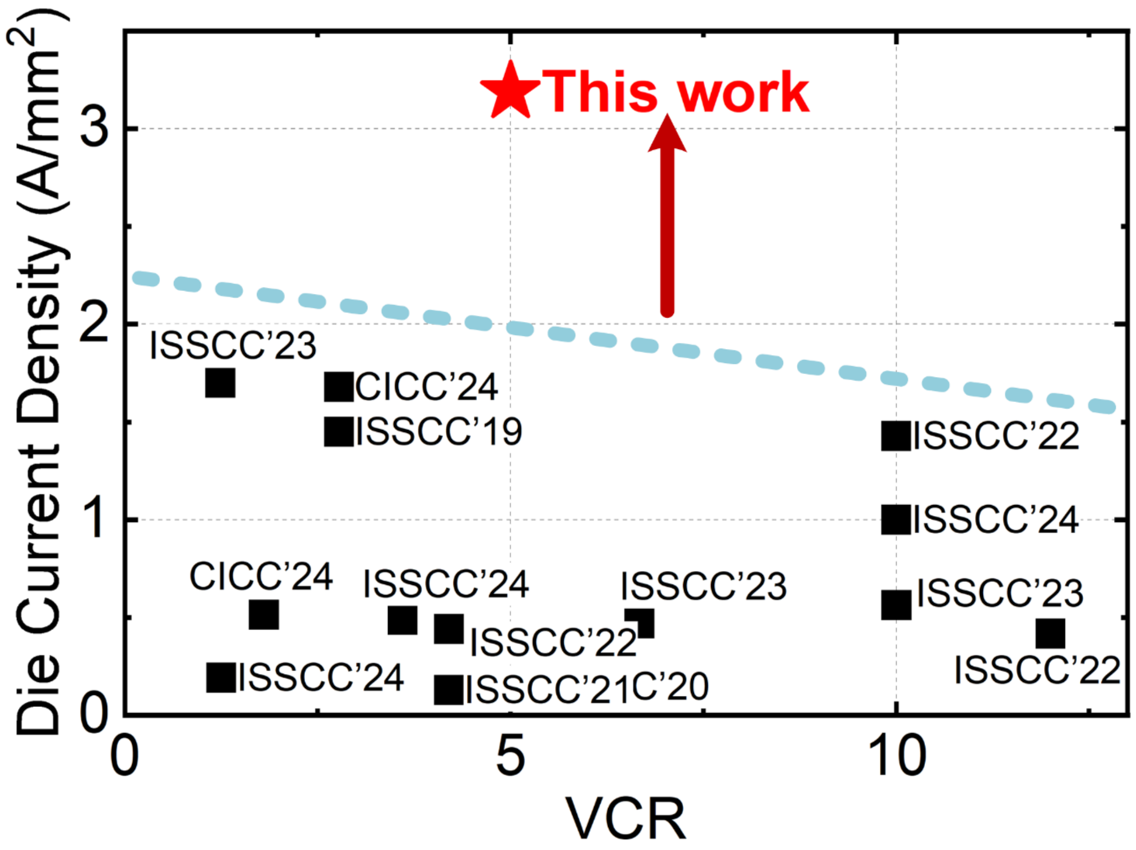
Measured Current Density & FoMs

- VPD tile achieves 3.2A/mm^2 and fast load transient FoM as 32ns

FoM Comparison



Current Density Comparison



Comparison with Prior Arts

	[1]	[9]	[8]	[12]	[11]	[10]	***This Work***
Topology	4L FCML	CCC	Buck	DSD	Dual-path	Dual-path	**5L FCML**
Process	22nm	180nm	180nm	180nm	180nm	180nm	**65nm**
Input Voltage	5V	12V	5V	12V/24V	12	2.7-4.2V	**3.8-5V**
Output Voltage	1.8V	0.9-1.8V	1.2-2.5V	1V	1-1.8V	3.4V	**0.6-1.2V**
Max Output Current	10A	4A	5A	4A	5A	2.5A	**10A**
Conduction Mode	CCM	CCM	CCM	CCM	CCM	CCM	**CCM/DCM**
Load Step	0→6A	0→4A	0→2A	0.1→3.1A	0.5→2.5A	0.5A	**0.2→10A**
V_{out} Undershoot	150mV	11mV	70mV	56mV	128mV	155mV	**68mV**
Settling Time	2.5µs	1.5µs	1.14µs	0.9µs	32µs	23µs	**3.3µs**
Peak Efficiency	90.5%	86.8% @1.2V	95.8% @1A	88.3%	93.7% @1.2V	98.6%@0.5A	**93%@1.2A**
Inductance	10nH	2×740nH (eff. val.)	880nH	2×1.8µH	1µH	2.2µH	**22nH**
Flying Cap	2×13.2µF	2×2.2µF	-	4.7µF	3×10µF 1×22µF	4.7µF	**6×4.7µF**
Output Decoupling	18µF	10.6µF	8µF (eff. val.)	10µF	22µF	10µF	**4.7µF**
Switching Frequency [MHz]	5	2	2.5	1	1	1	**1.6-32**
Die Area Current Density [A/mm²]	1.45†	1.43†	1.68†	0.417†	1.0†	1.7†	**3.2**
FOM [us]	0.45	0.292	0.28	0.187	1.408	3.1	**0.032**

Highest I_{Load}

Wide Load Range

Fully Integrated Vertical Power Delivery

Highest Density

$$FoM = C_L \times \frac{V_{droop}}{\Delta I_{Load,max}}$$

† The calculated die area does not include any passive components

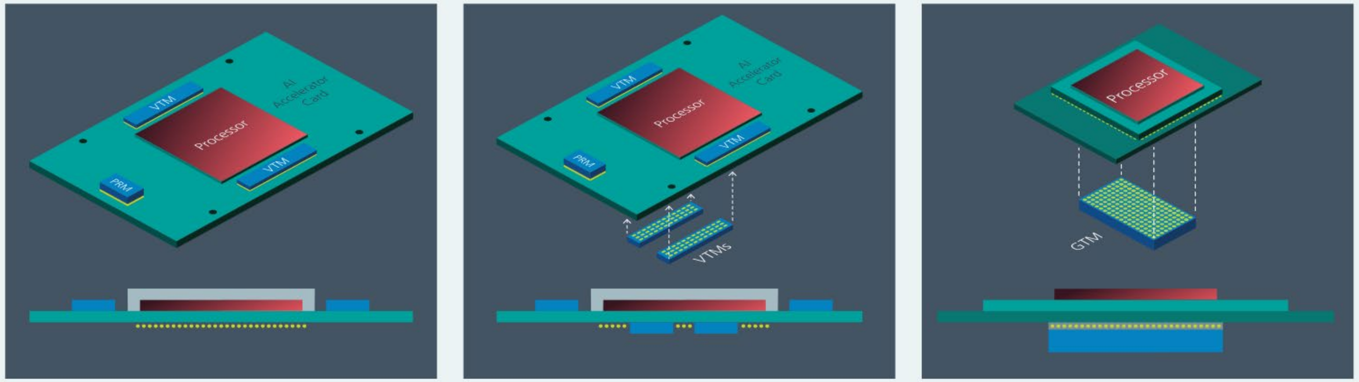
* The calculated die area includes passive component footprints

OUTLINE

- Next-Generation Datacenter and *Power Wall* Bottleneck
- Integrated Voltage Regulator Chiplet for 2.5D/3D Processors
- **Vertical Power Delivery (VPD)**
- Power Architecture for HPC Datacenter
- Conclusions

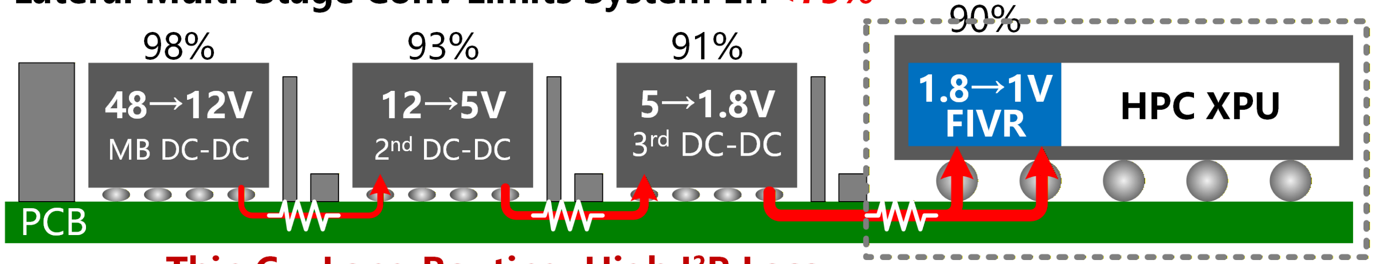
Conventional Power Management Solution for XPUs

- **The *Last Mile* of Power Delivery**
  - **Datacenter 48-1V regulation uses series 3-/4-stage arch, too many conversion stages & losses; limited PCB copper thickness & current capability → TDP & IPC Bottleneck**
- **High-Voltage Conversion reduces stages, increase efficiency**
  - **BUT needs higher breakdown voltage & disruptive wide bandgap semiconductor**
- **Reduce delivery distance & form factor → Enhance power density, reduce IR losses → Boost Efficiency**
  - **Replace Lateral Power Delivery (LPD) w/ emerging Vertical Power Delivery (VPD)**



Lateral Power Delivery	Lateral/Vertical Power Delivery	Vertical Power Delivery
Best suited for loads <800A	Best suited for loads <1200A	Best suited for loads >1200A
Current multipliers reduce PDN resistance	Further optimization of PDN resistance via	Lowest PDN resistance
SAC current multipliers are quiet in operation	VTMs on North and South side of the load	GTM directly injects current into load BGA
East and West side open for I/O proliferation	One or two VTMs on the bottom of the load	On-board bypass capacitors
		Maps VTM pinouts to load pinouts
<70μΩ of PDN resistance	<20μΩ of PDN resistance	<10μΩ of PDN resistance
Top side cooling	Top and bottom side cooling	Top and bottom side cooling

Lateral Multi-Stage Conv Limits System Eff<75%

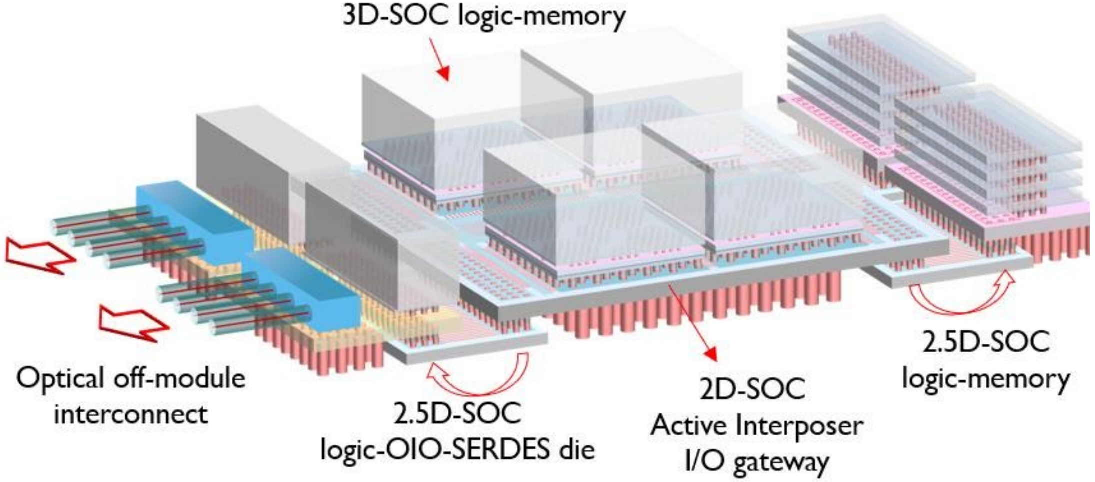


Thin Cu, Long Routing, High I^2R Loss

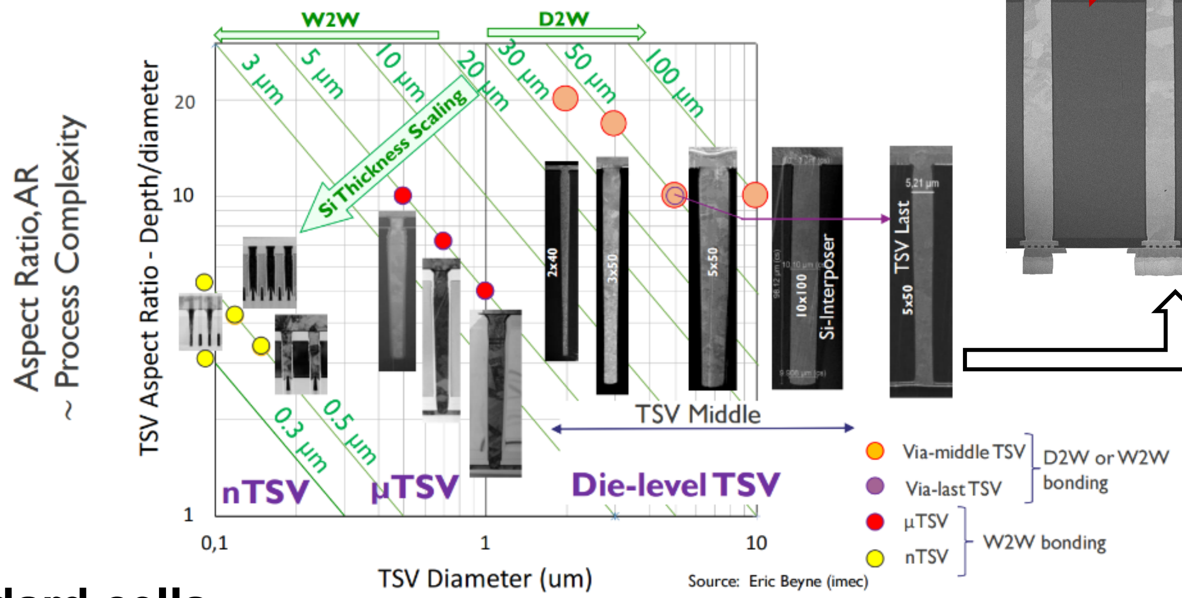
Integration of Backside Power Delivery Networks (BSPDN)

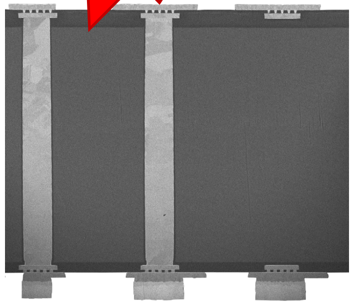
Not VPD

High-Bandwidth 3D Integration



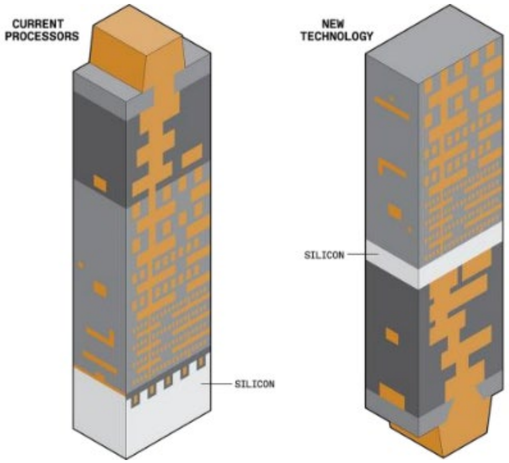
3D Technology: TSVs

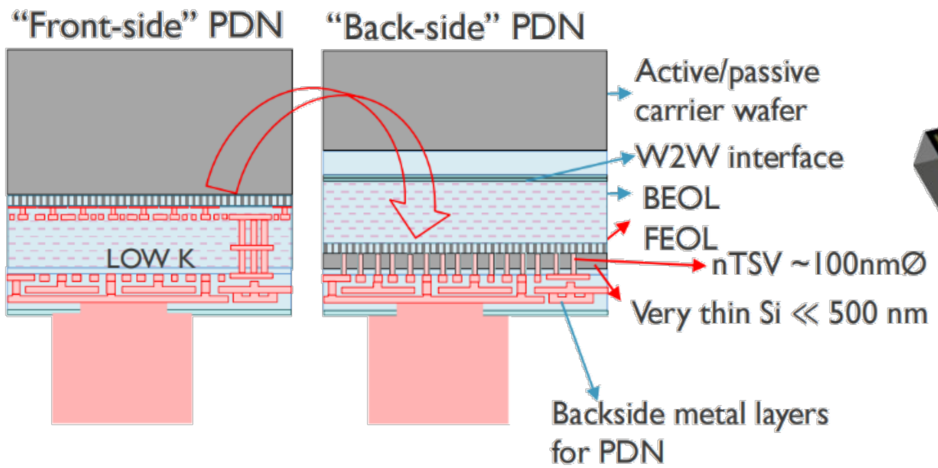




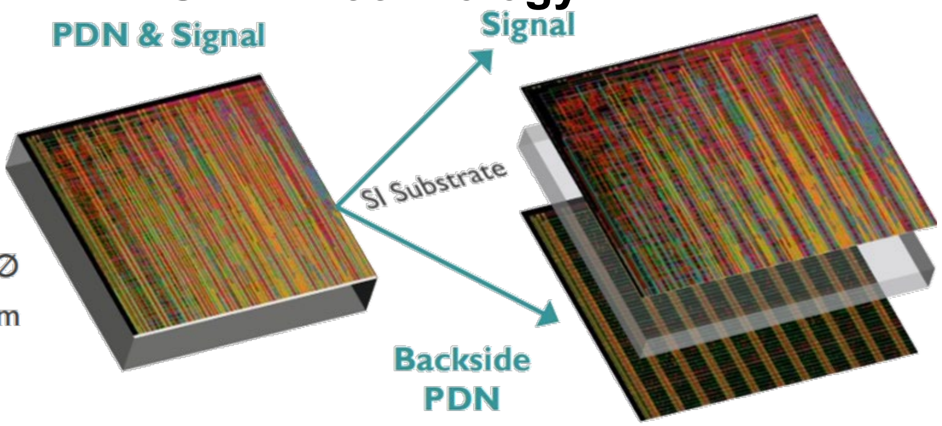
Silicon Thinning to Reduce Aspect Ratio (AR)

Backside BEOL for direct power delivery to standard cells





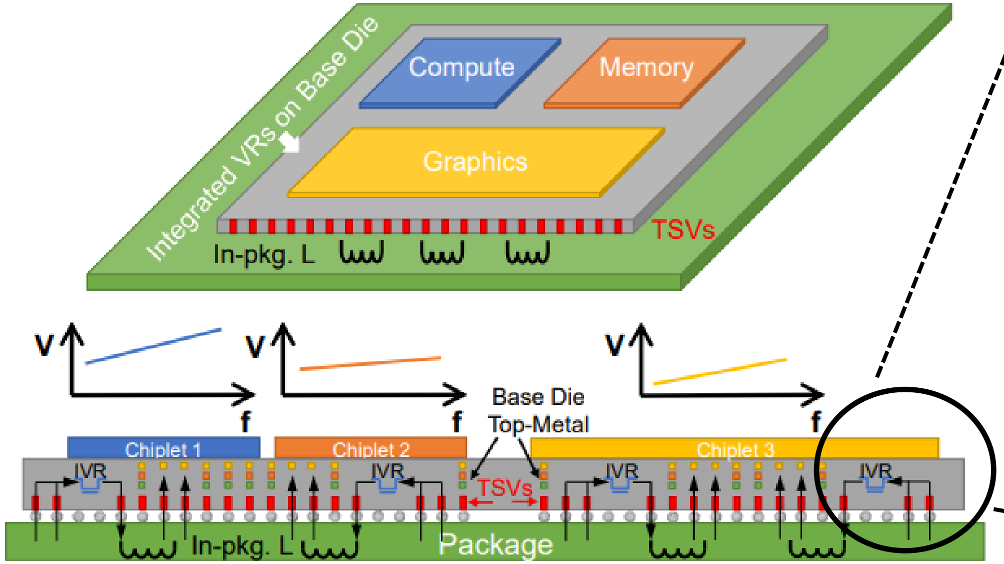
BSPDN Technology

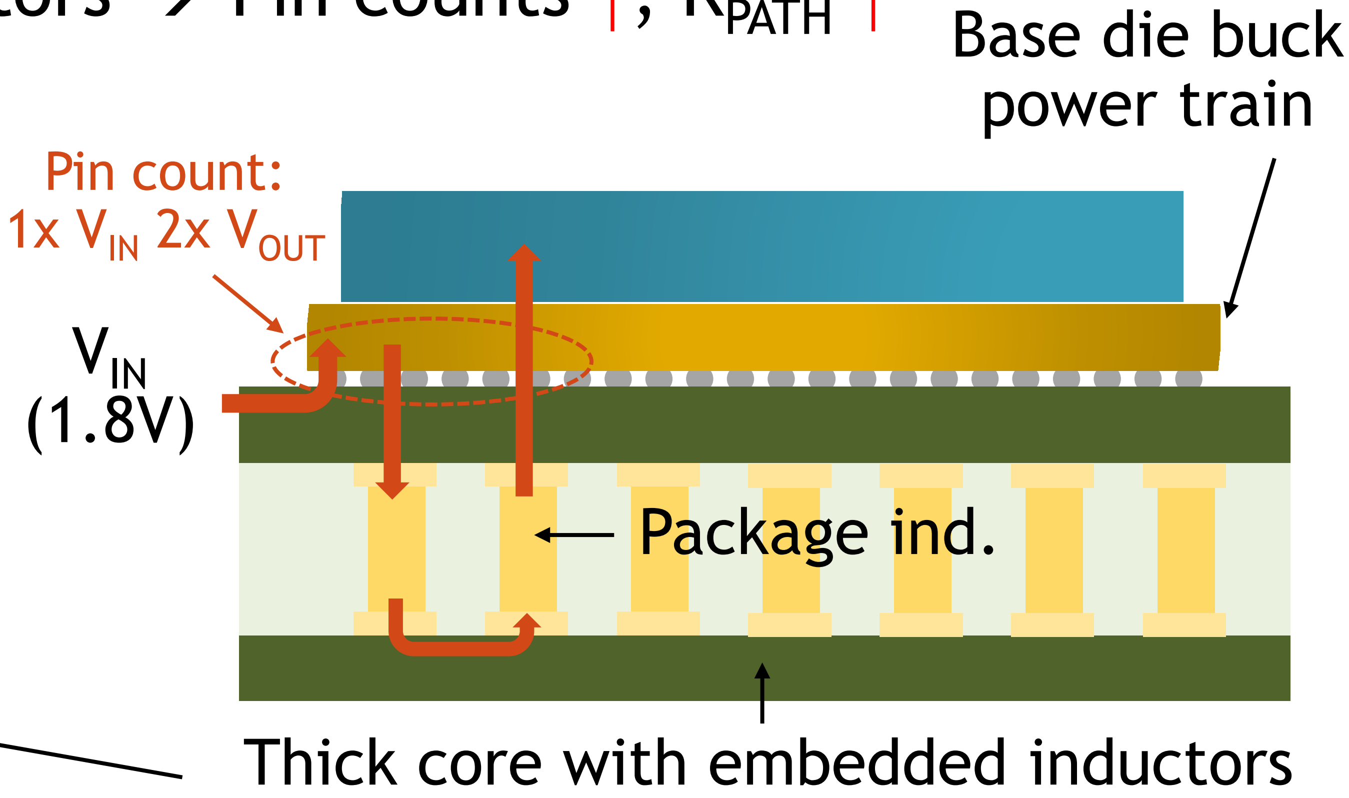


Prior Works

- Buck Converters as Base Die Voltage Regulators
  - – Package inductors required → Z-height ↑
  - – Extra pins to access inductors → Pin counts ↑, R_{PATH} ↑

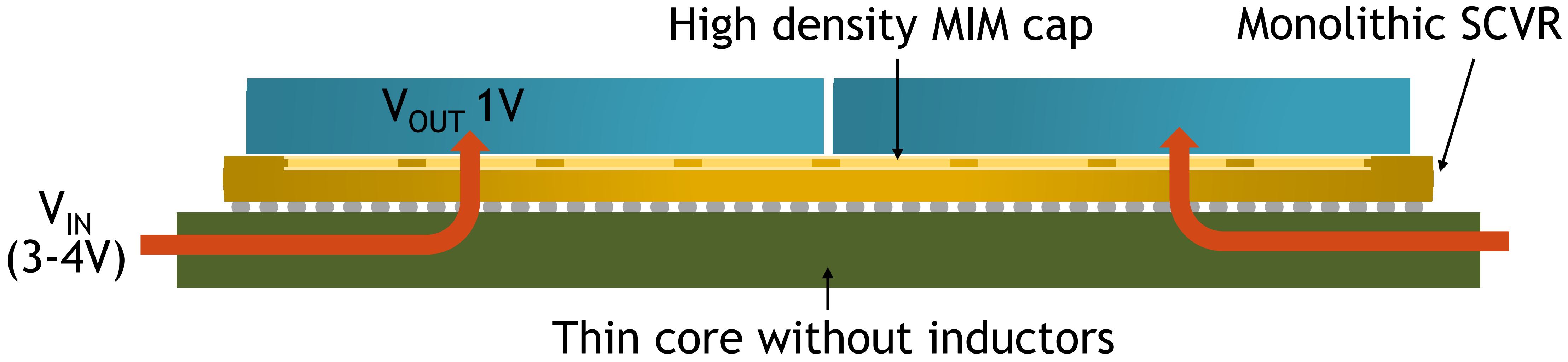
N. Desai, VLSI 2022. 85% peak eff.





Proposed Solution

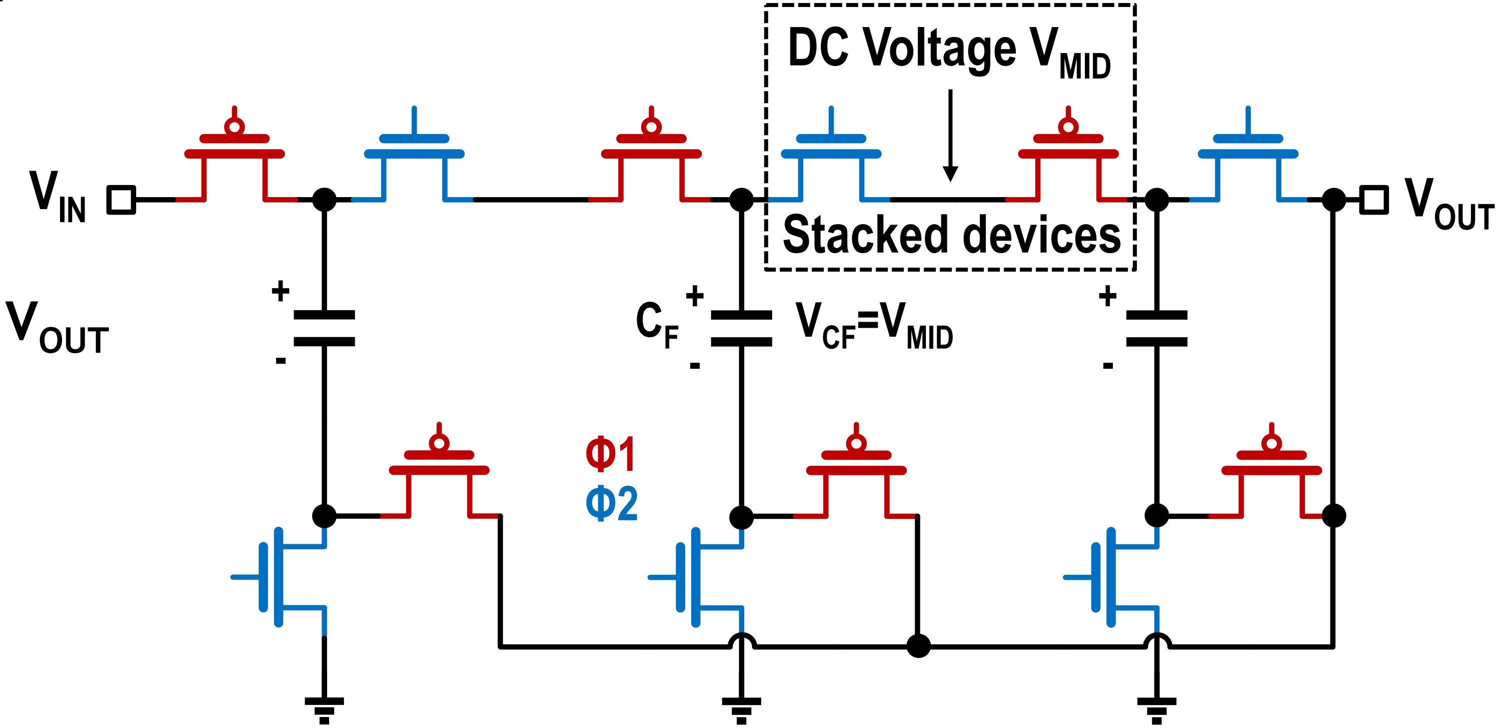
- Monolithic Switched Capacitor Converters as Base Die VRs
  - On chip MIM capacitors → Z-height ↓, Pin counts ↓, R_{PATH} ↓
  - Modular design for both V_{IN} and I_{OUT} → Scalability ↑
  - High V_{IN}, high efficiency & high power density.



Proposed 3:1 and 4:1 MMSCVR

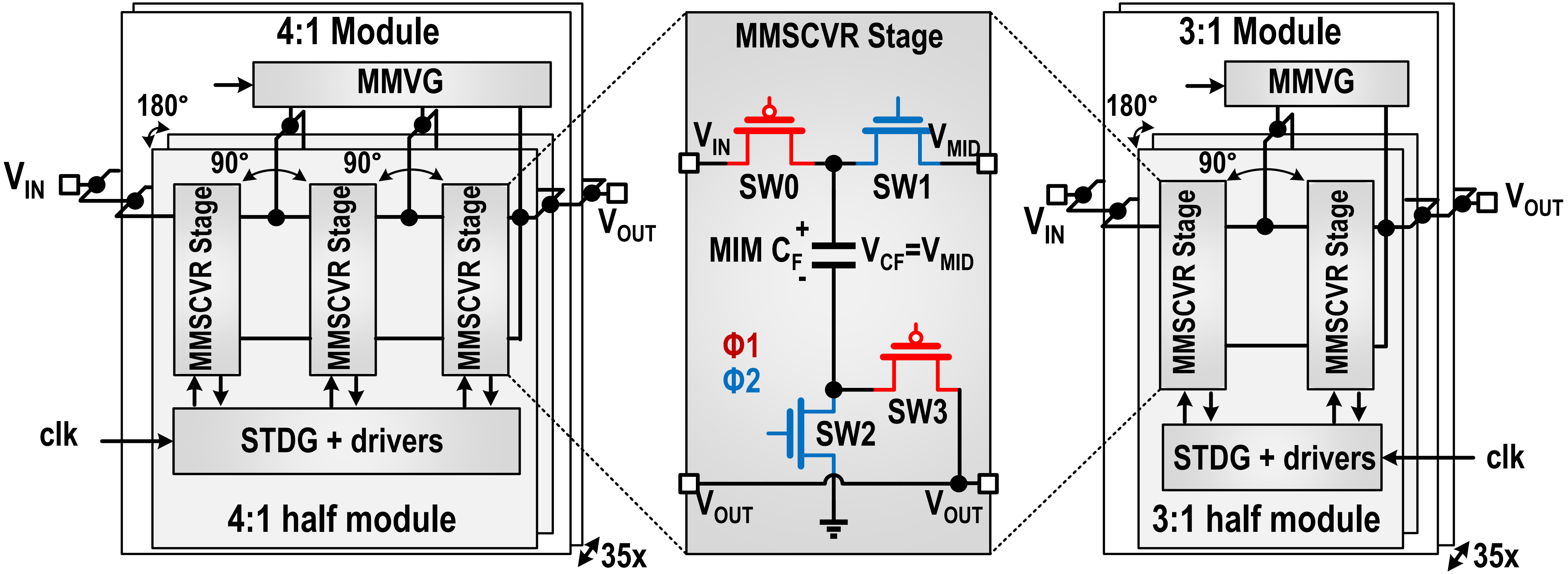
- Multi-Stage Modular Switched Capacitor Voltage Regulator (MMSCVR)
  - Using Dickson topology for excellent scalability and efficiency.
  - DC voltage V_{MID} between stacked devices offers convenience for driving and startup.

All switch stress = V_{OUT}



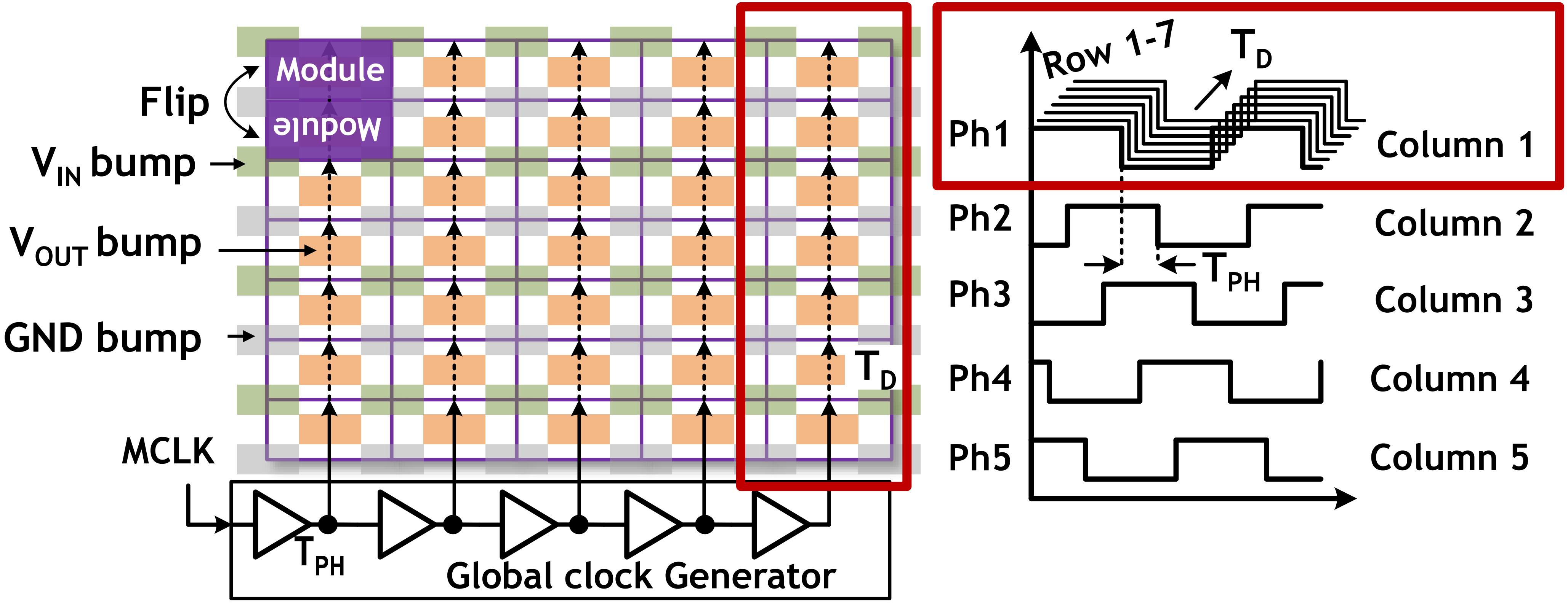
Proposed 3:1 and 4:1 MMSCVR

- Multi-Stage Modular Switched Capacitor Voltage Regulator (MMSCVR)
  - Self-timed deadtime generator (STDG) for deadtime control.
  - Multi-level mid-rail generator (MMVG) for startup.



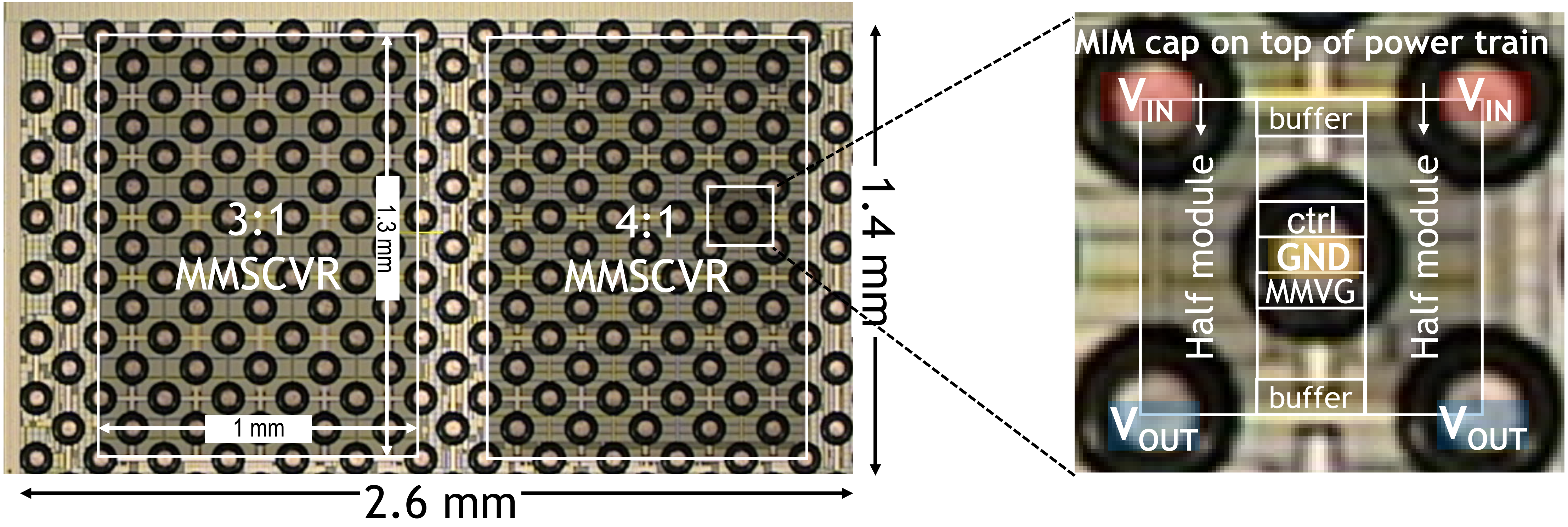
System and Clock Generating

- 7x5 modules form the power plane.
- Programmable delay T_{PH} for column phase-interleaving.
- Natural propagating delay T_D for row phase-interleaving.



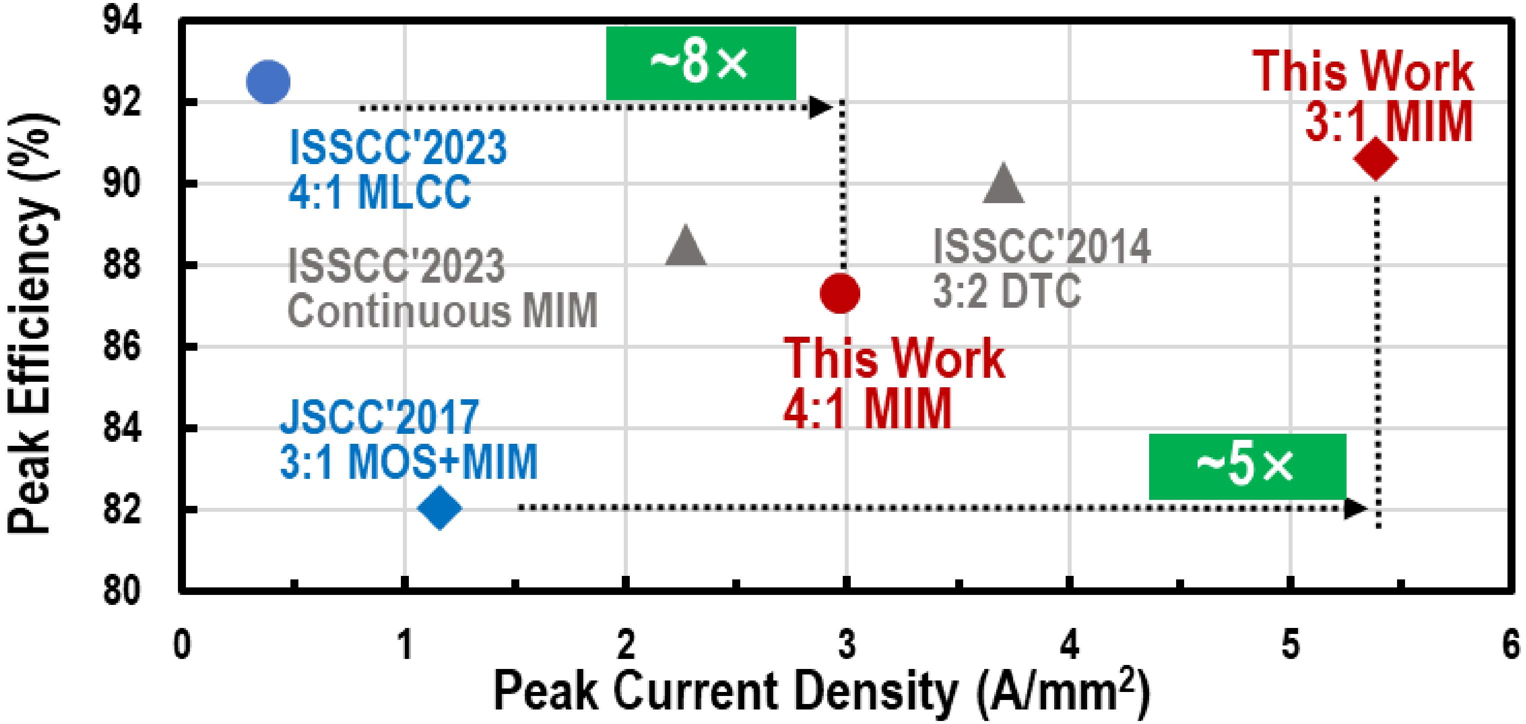
System Implementation

- Fabricated in 16nm CMOS process.
- 3:1 and 4:1 MMSCVR: 1mm x 1.3mm.
- MMSCVR module design.



Conclusion

- Multi-stage Modular SCVR in 16nm technology
  - Fully monolithic solution tailored for future vertical power delivery.
  - Scalable design on both V_{IN} and I_{OUT}.
  - Highest efficiency and density among state-of-the-art.



OUTLINE

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- Vertical Power Delivery (VPD)
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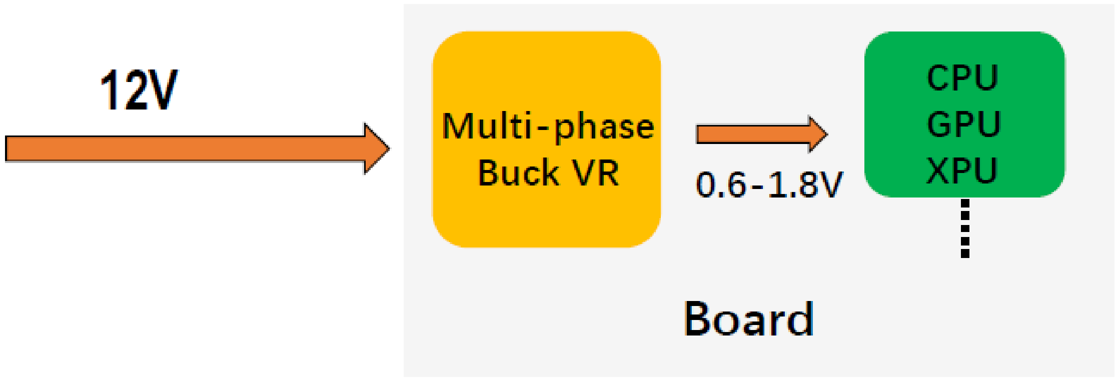
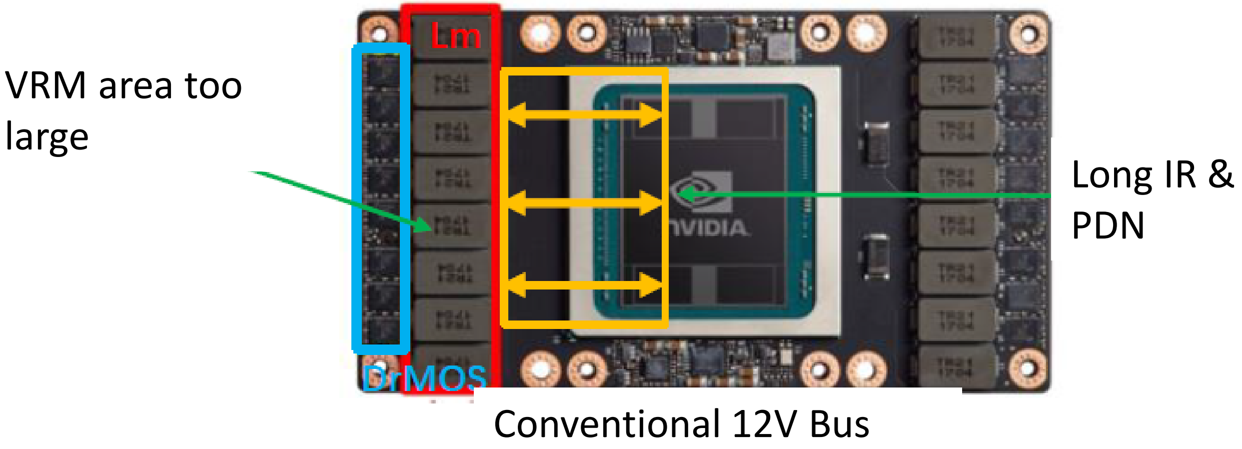
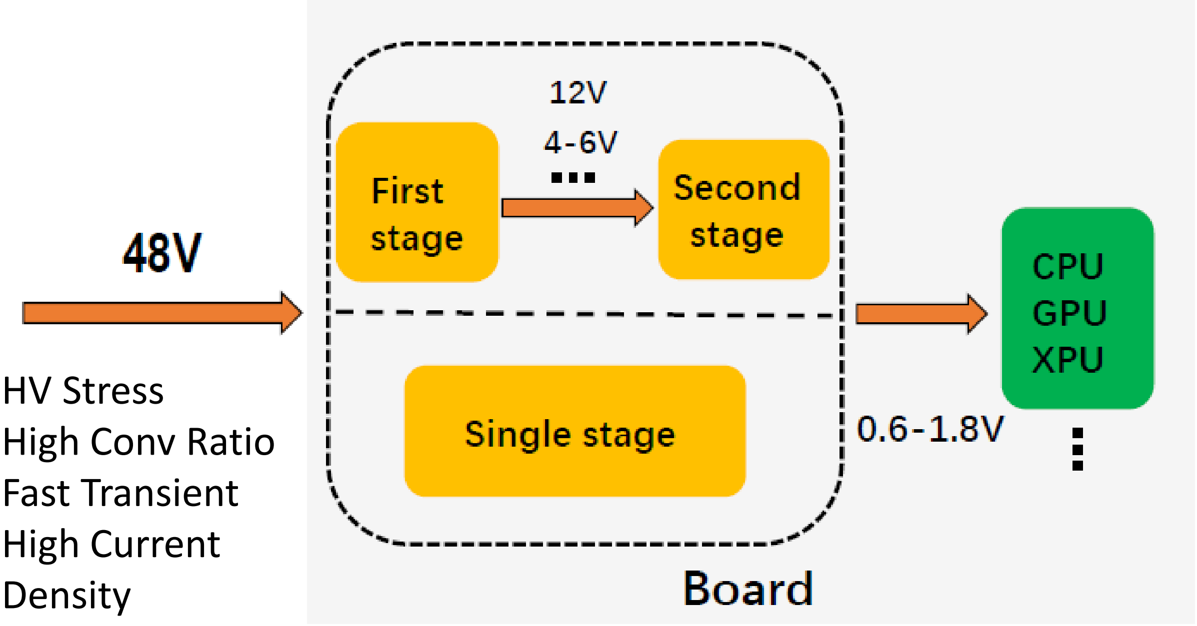
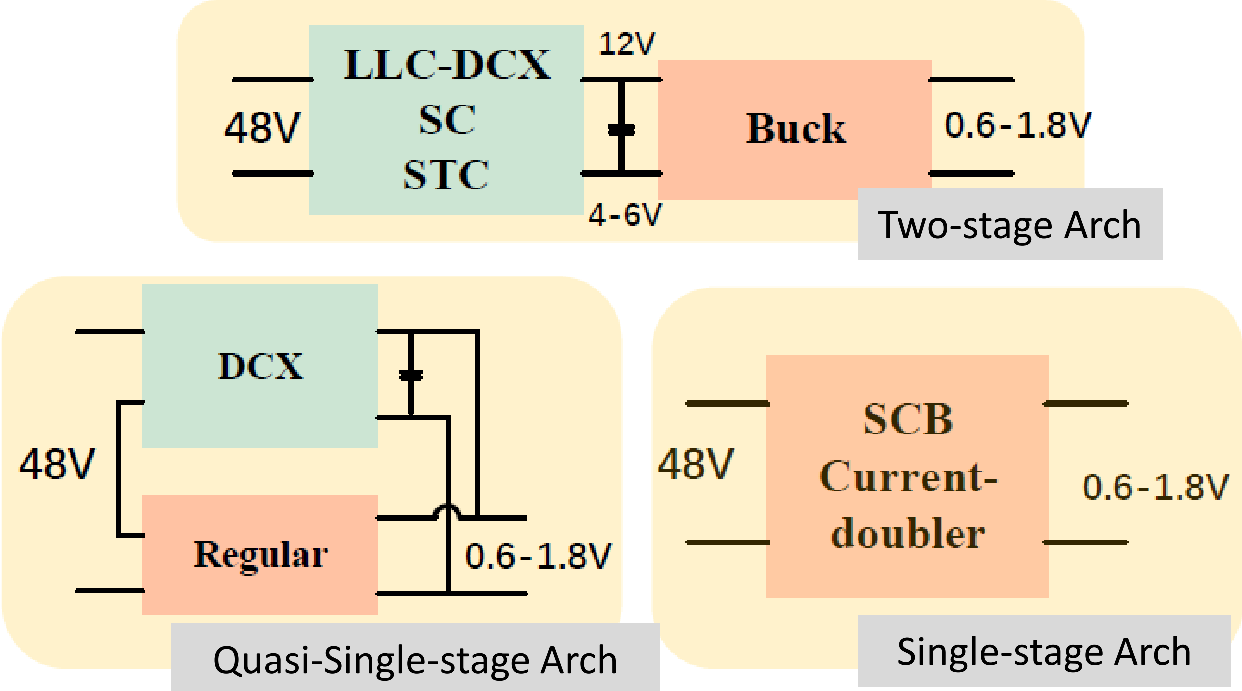
48V–1V Multiphase Power Delivery for XPUs



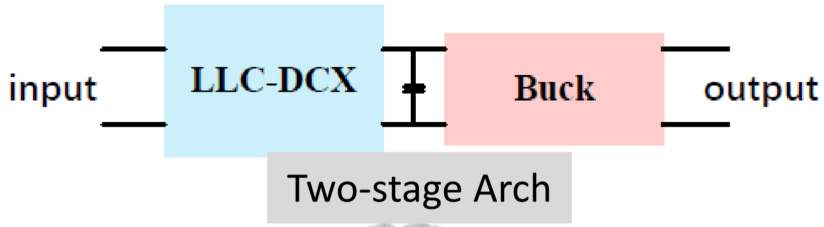


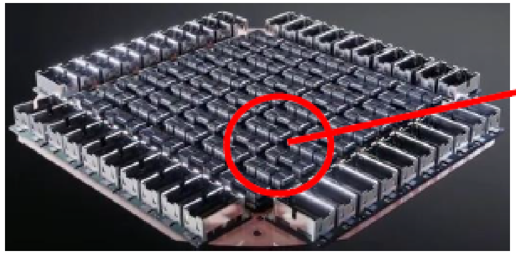


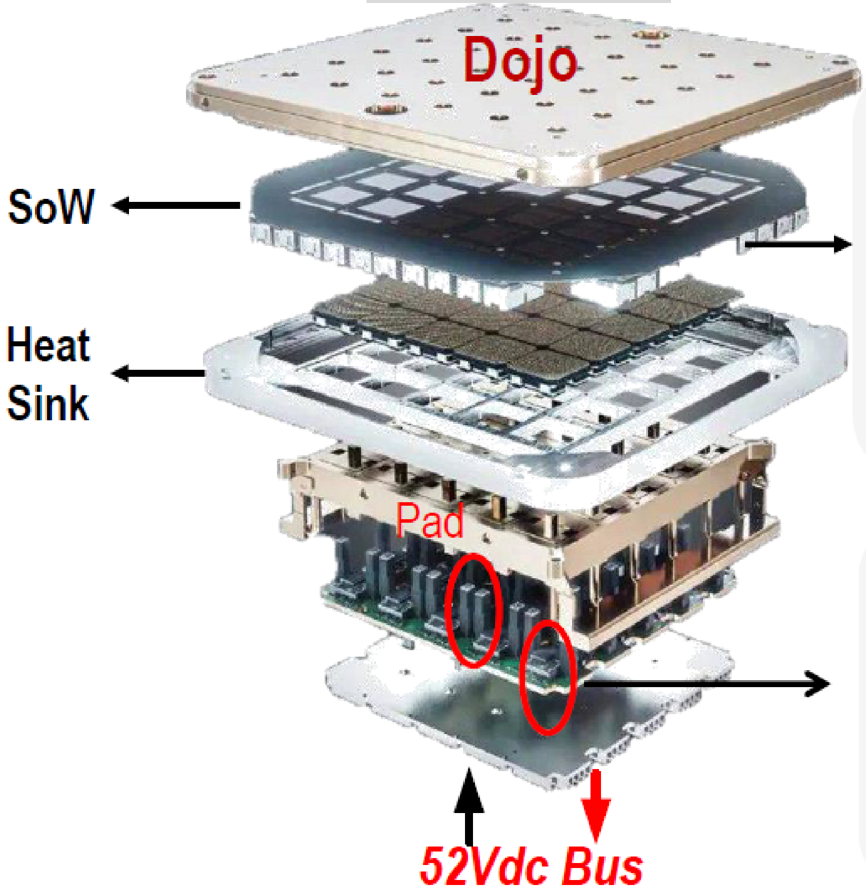
- HV Stress
- High Conv Ratio
- Fast Transient
- High Current Density

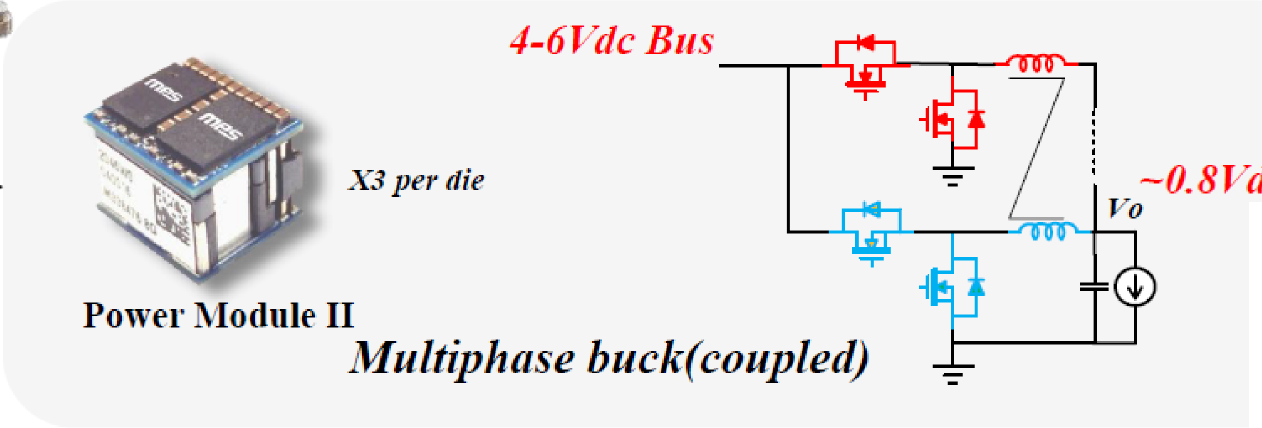


Tesla's Dojo HPC System: 52V, 25 Units, 15 kW

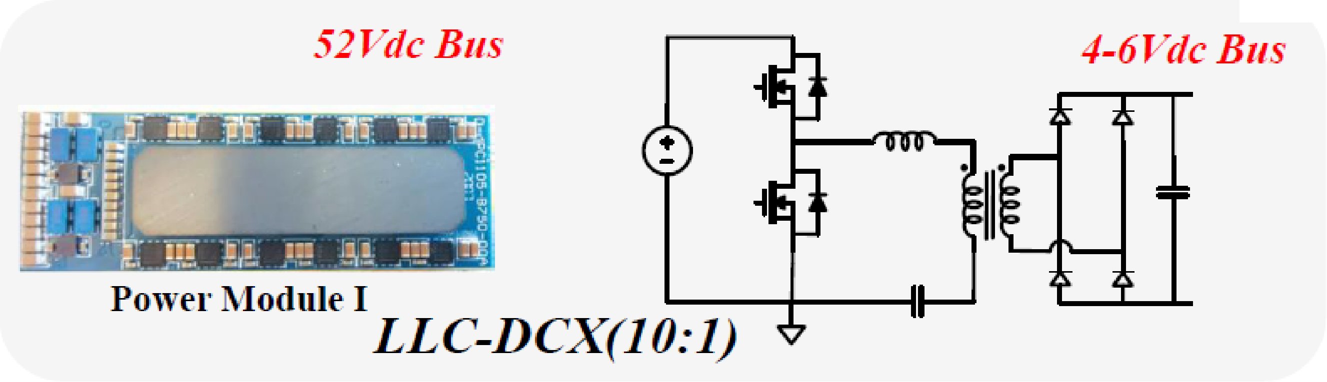




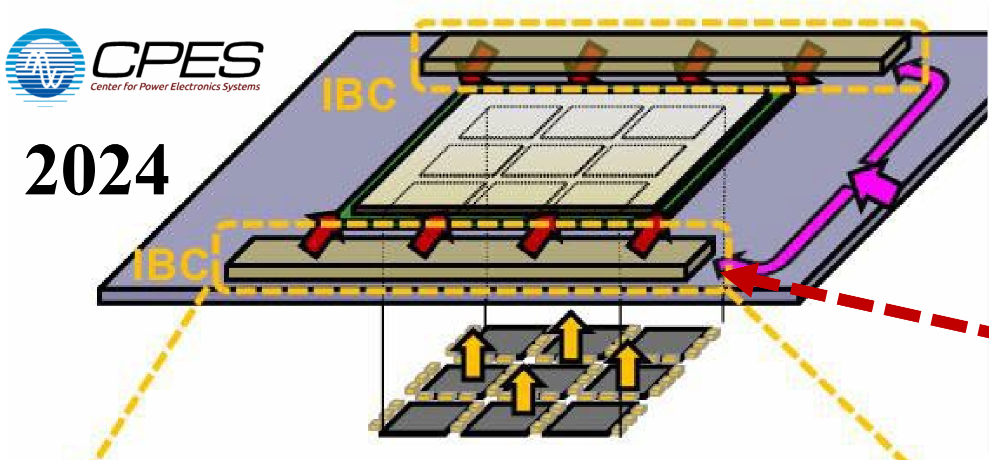


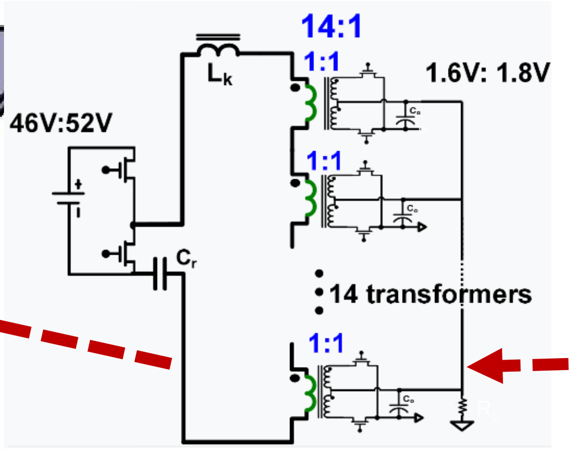


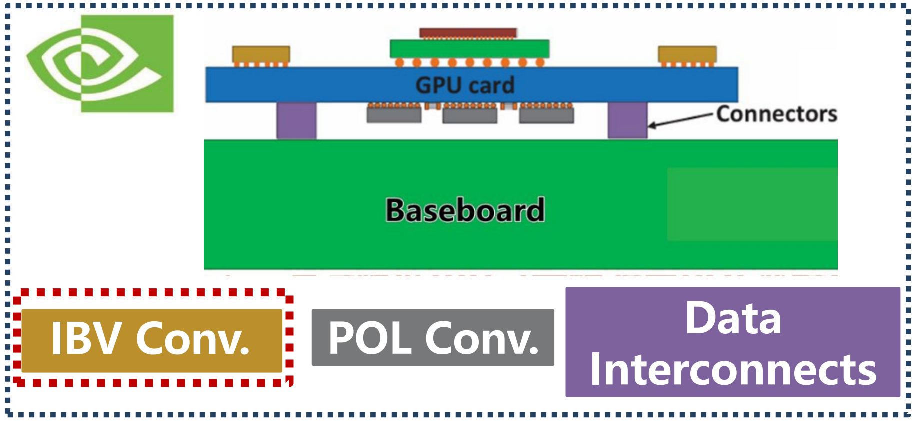
- VPD shorter IR
- Compatible w/ 12V IBC – **Good**/**Bad**
- Worse loss & size of former stage

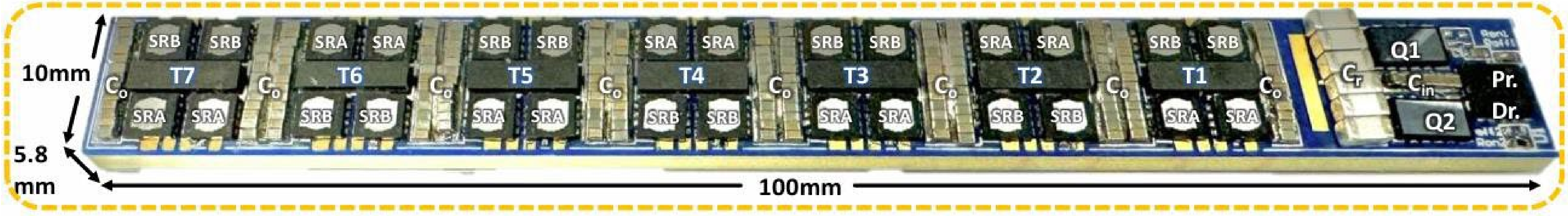


kW-Level Power Delivery (48V→1.8V→1V): A Power Electronics View









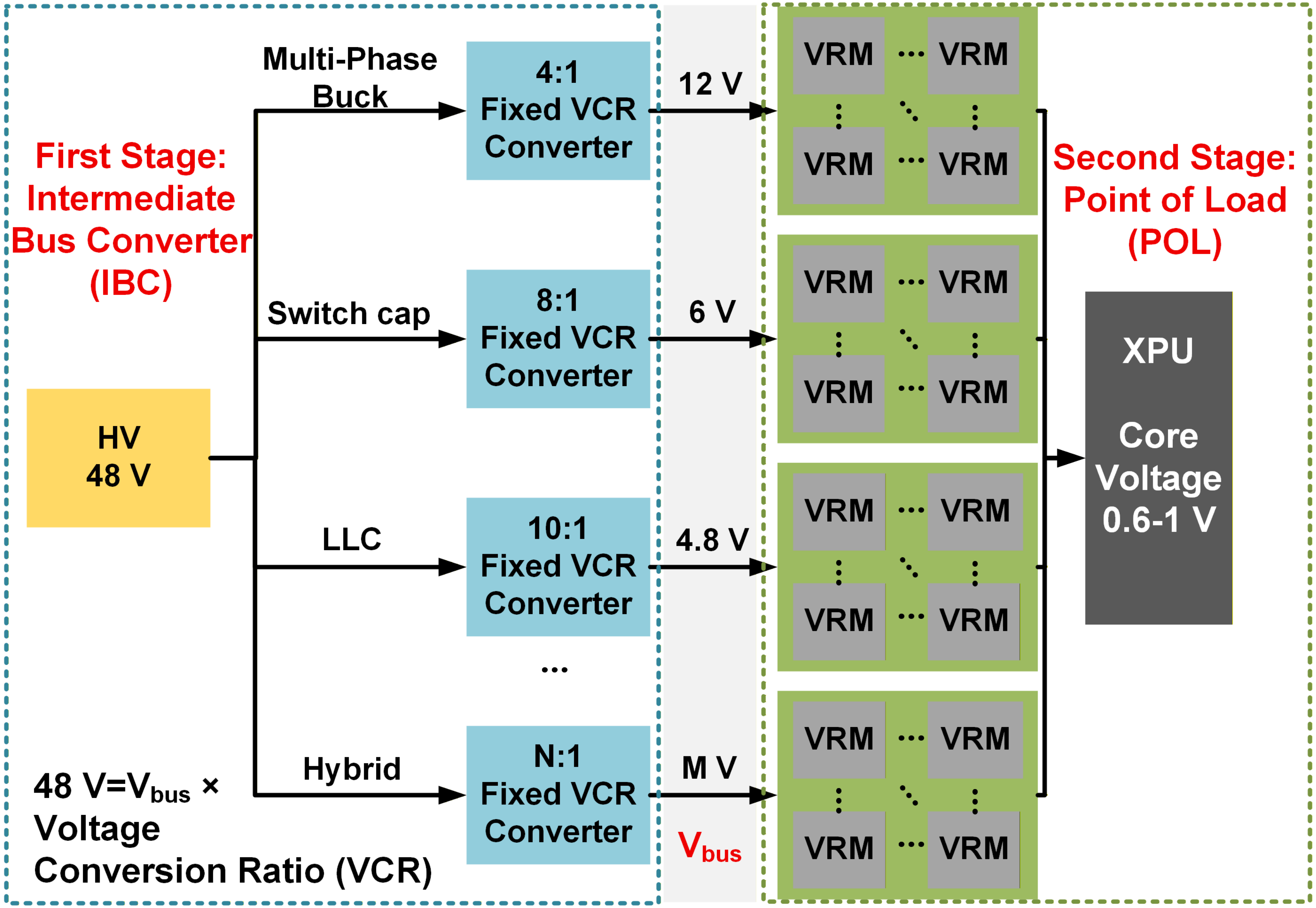
48V/1.8V LLC Front-End Converter: 95% Peak Eff, 2.4 kW/in³

Lower Intermediate Bus for Performance Improvement

Schematic of the HPC Power-delivery Architecture

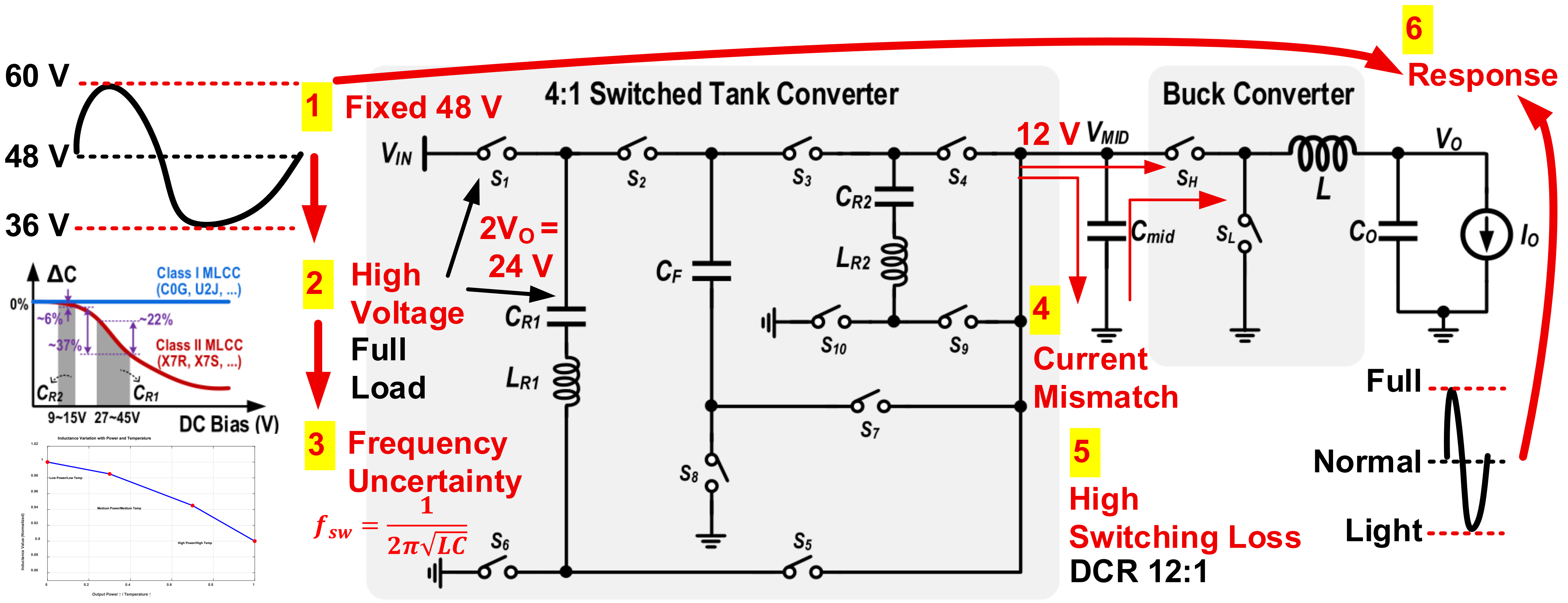
Datacenter requirements

- High efficiency
- High power density
- High voltage conversion ratio
- Wide input range
- Load transient response
- Protection features



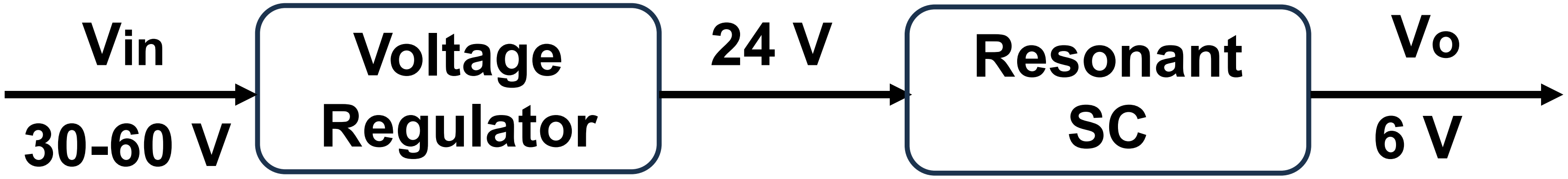
Conventional Schematic of the Converter——Google TPU

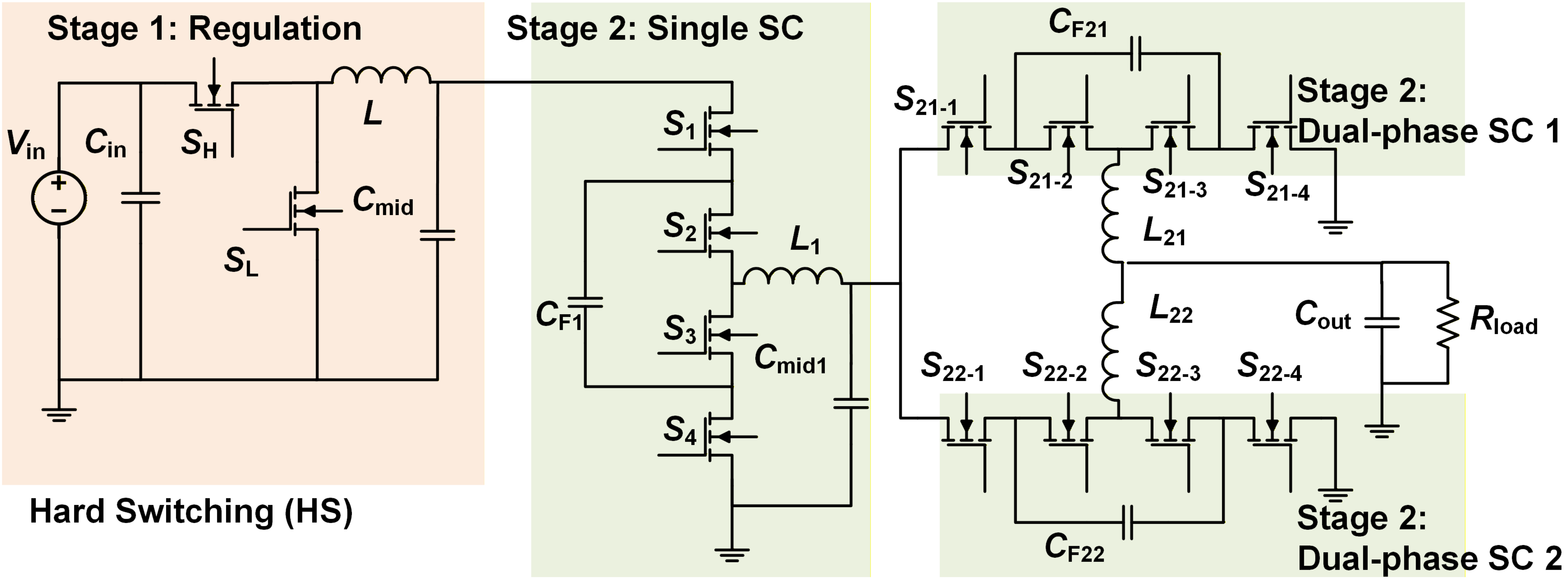
Mismatch between wide input voltage range and high performance



Proposed Regulated Resonant Switched-capacitor

Proposed circuit

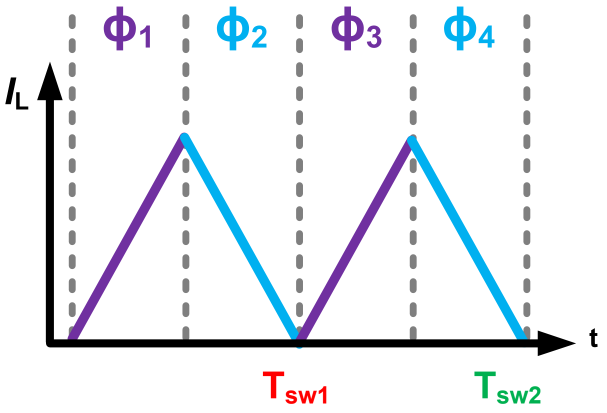




Proposed Circuit with 2:1 Ratio

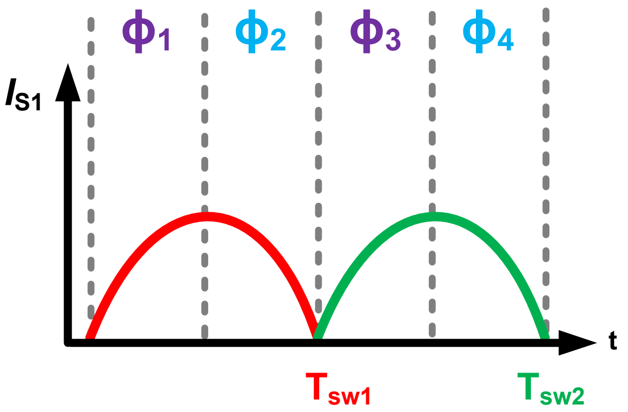
Working principle of the proposed R²SC converter

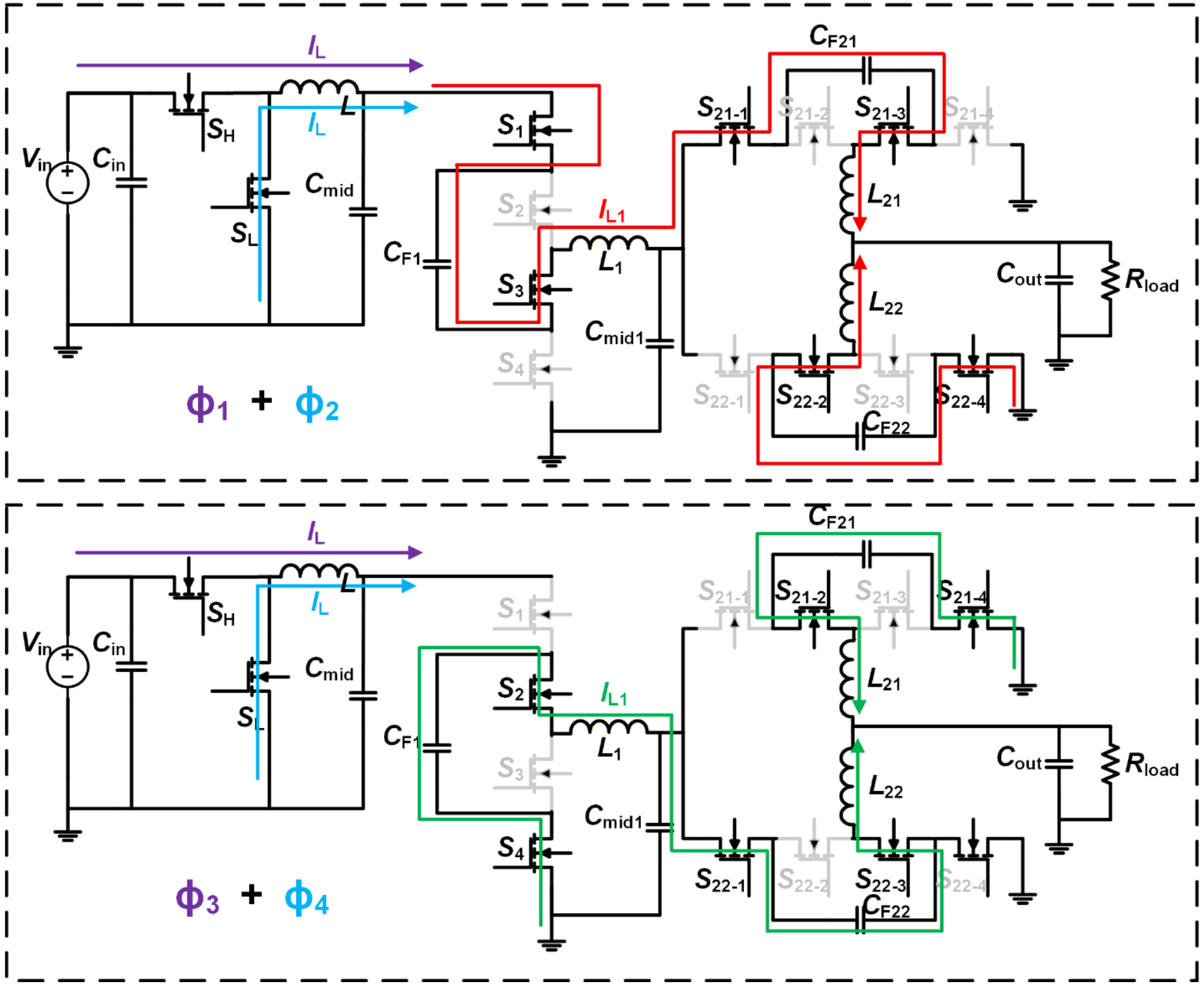
Stage1



Stage2

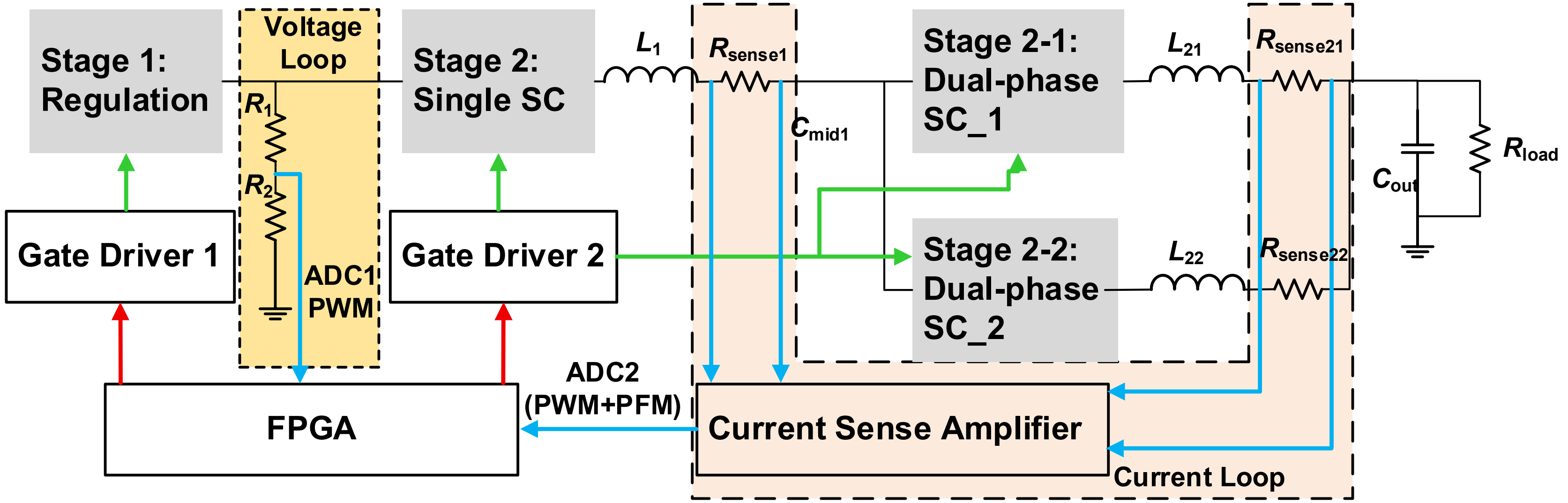
$T_{sw2} = 2T_{sw1}$



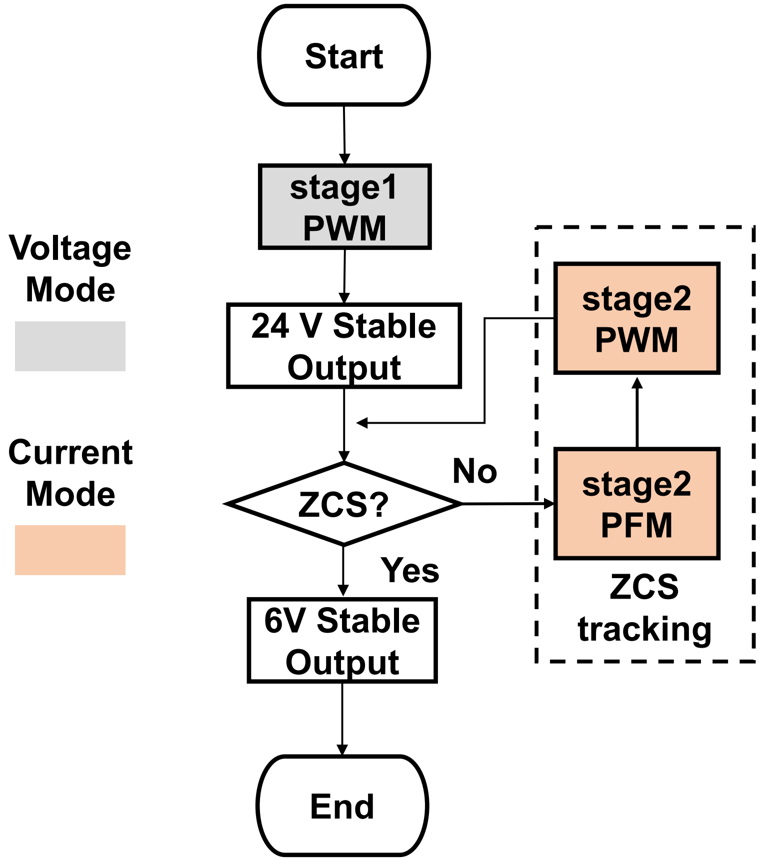


Unified Hard–Soft Switching Control

System architecture of the R²SC converter



Processes of unified control

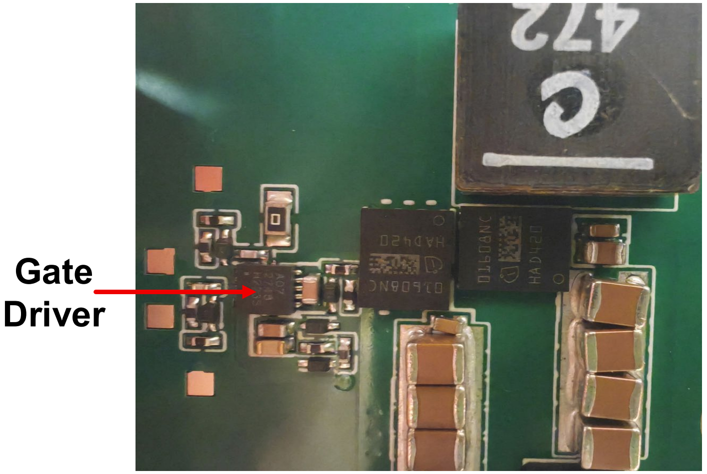


Implementation of the Converter

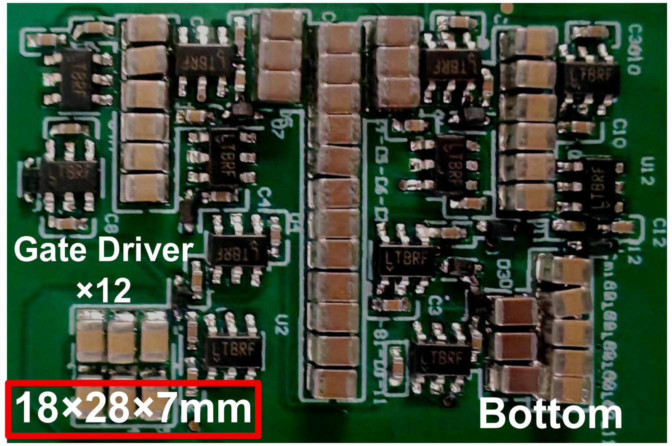
Main Component List of R²SC

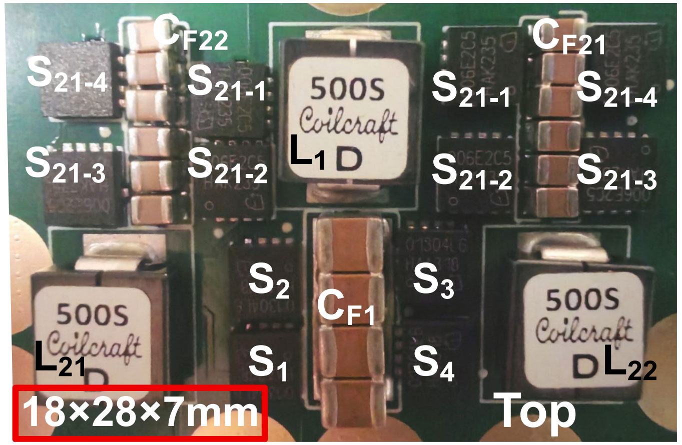
Components	Information	Parameters
Gate driver 1	Analog Devices, LTC4440-5	80 V, high-side
Gate driver 2	Infineon, 2ED2748S01G	140 V, high-side
Stage 1 MOS	Infineon, IQD016N08NM5CG	80 V, 1.57 mΩ
Stage 2 MOS 1	Infineon, IQE006NE2LM5CG	25 V, 0.65 mΩ
Stage 2 MOS 2	Infineon, IQE006NE2LM5CG	25 V, 0.65 mΩ
CF1	Murata, GRM21BR61E226ME	22 μF, 25 V
CF2	Murata, GRM21BR61E226ME	22 μF, 25 V

Stage 1



Stage 2

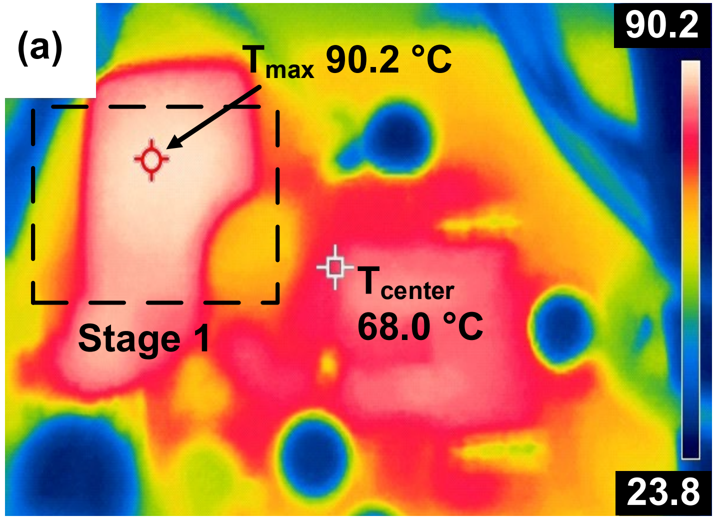




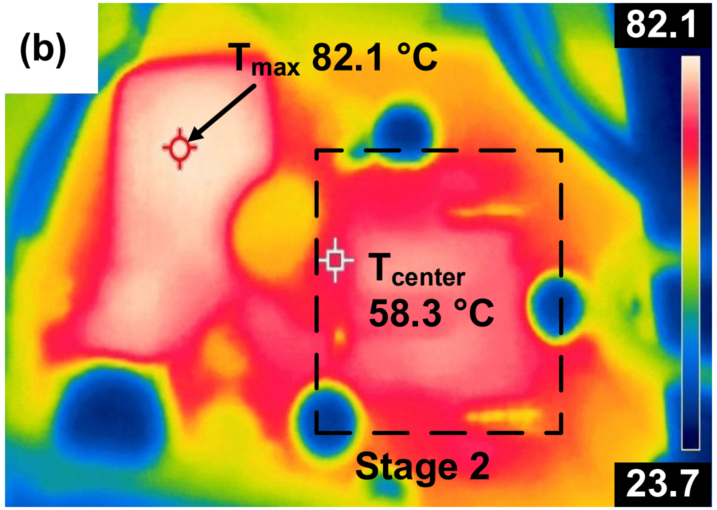
Experimental Results

Thermal results

Before UHSC strategy



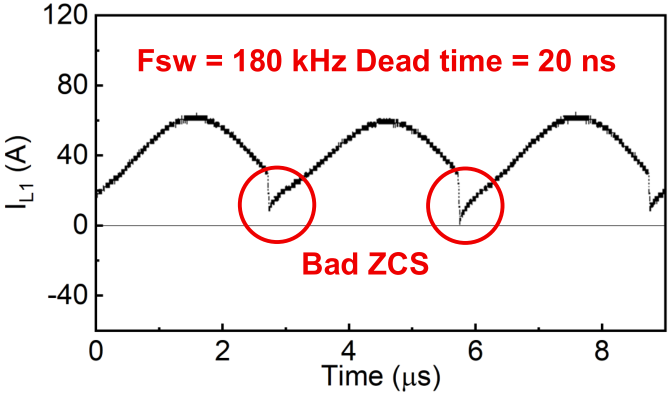
After UHSC strategy



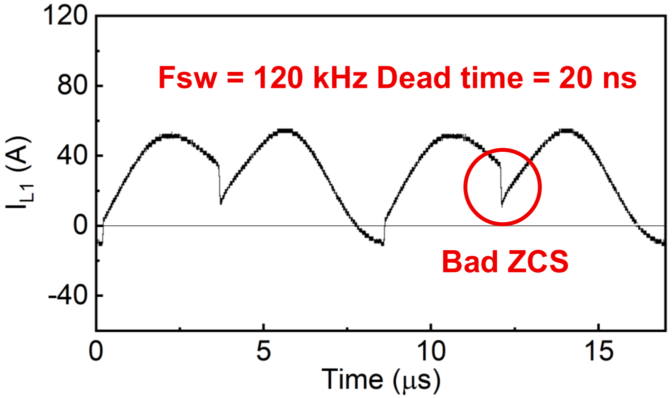
Thermal performance with 7200 RPM (fan cooling)

Waveforms depicting the inductor current

Before UHSC strategy

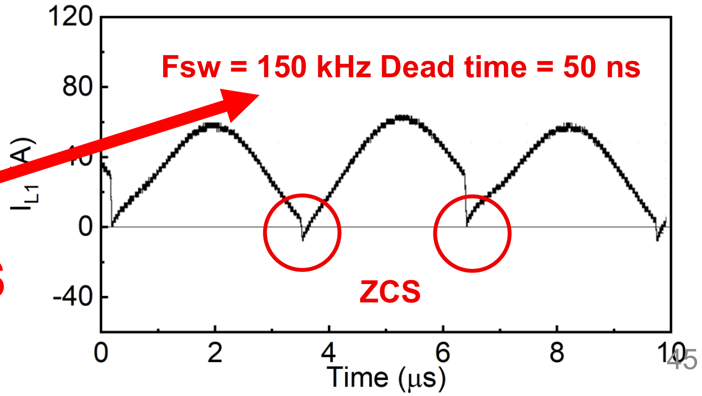


During UHSC strategy



After UHSC strategy

Best freq to achieve the ZCS



Summary and Comparison

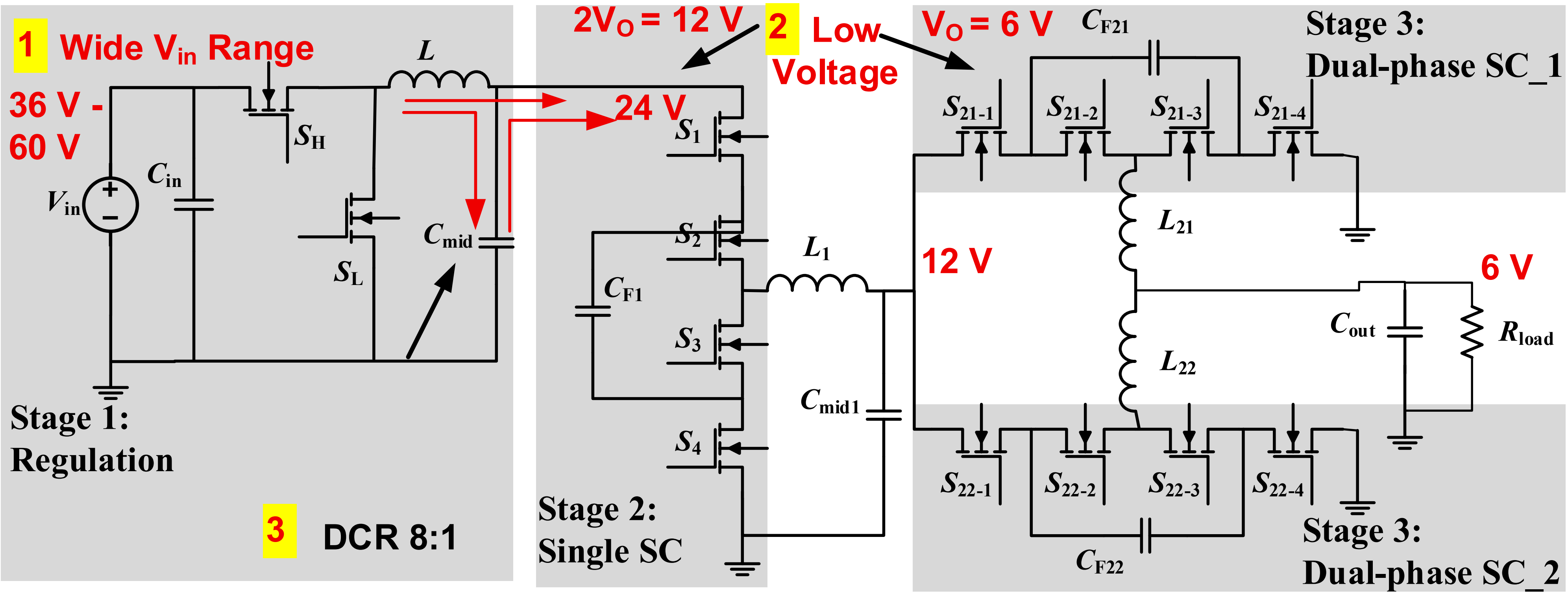
Performance comparison between this work and existing works

Architecture	Unregulated			Regulated			
Topology	CaSP	STC	LLC	DPHD	DSDSTC	ReSC	This Work R^2SC
VCR	8:1	4:1	8:1	Continuous			
Vin	48 V	48 V	48 V	36-65 V	48 V	36-60 V	30-60 V
Vout	6 V	12 V	6 V	1-2 V	1-3.3 V	3.3 V/5 V	6 V
Max Iout	30 A	50 A	150 A	30 A	50 A	30 A	100 A
Max Pout	180 W	600 W	900 W	60 W	165 W	150 W	600 W
Peak Eff	98.4%	97.2%	98.1%	92.7%	94.7%	95.6%	95.6%
Density	NA	513 W/in³	1200 W/in³	451 W/in³	595 W/in³	567 W/in³	1339 W/in³

Conclusion

Regulated Resonant SC Converter with Unified Hard-Soft Switching Control

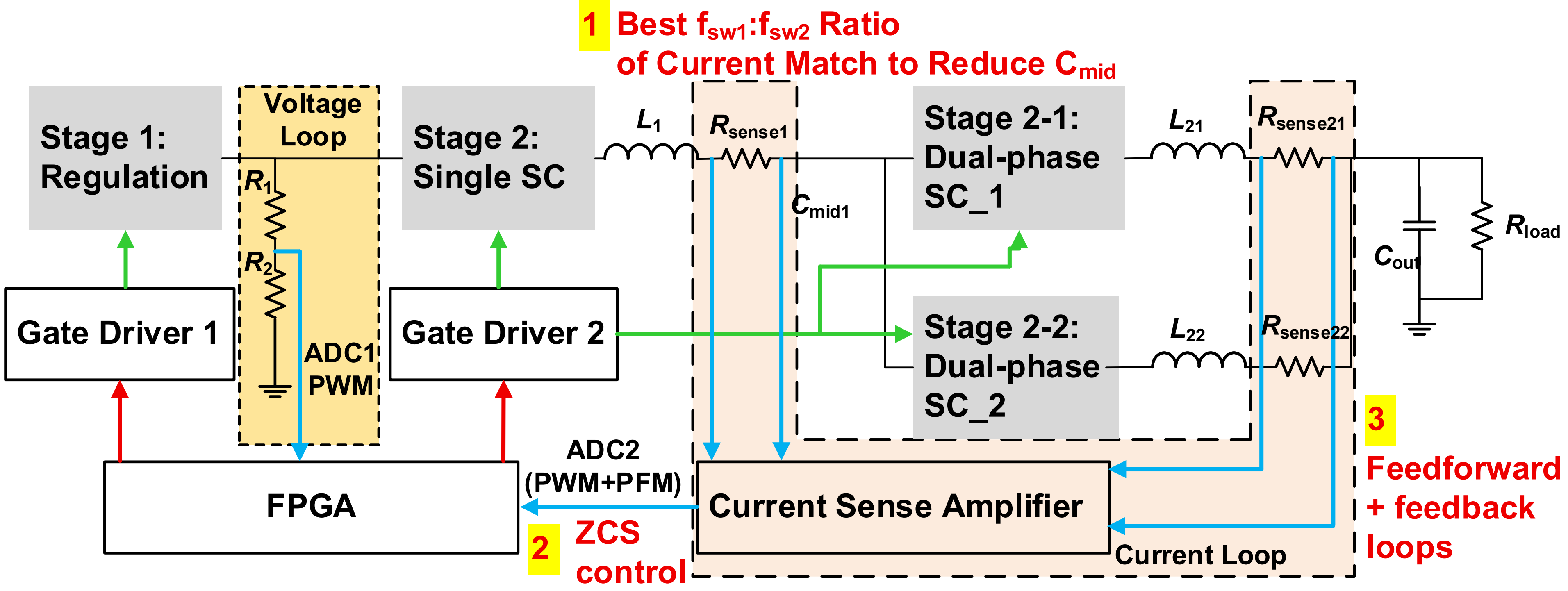
1. Front-stage wide voltage input
2. ReSC second stage: Hign quality DC bias immunity in low voltage range, reduced capacitor count, high power density
3. Low front-stage loss, reduced step-down burden on second stage (8:1)



Conclusion

Regulated Resonant SC Converter with Unified Hard-Soft Switching Control

1. Optimized switching ratio to minimize intermediate capacitor
2. Considering voltage-dependent capacitance and temperature-dependent inductance, zero-crossing detection ensures ZCS, real-time frequency optimization for higher efficiency
3. Feedforward + feedback loops for fast response to input/output transients



Conclusions

- **Next-Generation Datacenter and *Power Wall* Bottleneck**
  - Kilowatt-level power for each XPU, last mile vertical power delivery (VPD)
- **High Speed & High Voltage Trends for HPC Processors**
  - Novel architecture with new materials, devices & packaging technologies

THANK YOU!

清華大學
Tsinghua University